Phase 1 Technical Report: Problem Formulation, Data Understanding, and Quality Audit

NYC Yellow Taxi Hourly Demand Forecasting (2025)

Data Analysis Course Project (Fall 2026 / 1405)
Track A: NYC TLC Structured Time-Series Analysis

October 27, 2026

Abstract

This report presents Phase 1 of the NYC Yellow Taxi demand forecasting project for the full calendar year 2025. Leveraging a local DuckDB analytical engine operating directly over 12 monthly Parquet files (48.7 million trip records), we execute a comprehensive data structure and data quality audit. We formalize the operational decision problem—assisting fleet operators in pre-positioning vehicles across 265 taxi zones establishing the spatio-temporal aggregation unit $(z, t) \in \text{Zone} \times \text{Hour}$.

1 Problem Definition and Motivation

Taxicab demand in dense urban environments exhibits high spatial variance and non-stationary temporal dynamics.The core motivation of this project is to assist vehicle fleet managers in dynamically pre-positioning taxicabs across official NYC Taxi and Limousine Commission (TLC) zones. By accurately forecasting hourly trip originations per zone, dispatchers can minimize idle cruising time, reduce passenger wait times.

2 Related Work

Short-term urban passenger demand forecasting traditionally relies on historical rolling averages and Auto-Regressive Integrated Moving Average (ARIMA) models. Recent advances employ non-linear gradient-boosted decision trees (e.g., LightGBM, XGBoost) and spatio-temporal graph neural networks. In the context of NYC TLC datasets, prior literature emphasizes the necessity of strictly preserving temporal order during model validation to avoid look-ahead bias and spurious performance metrics.

3 Data and Data Quality

3.1 Dataset Overview

The primary dataset consists of the official NYC Yellow Taxi trip records for the entire calendar year 2025 (January 1, 2025 – December 31, 2025), partitioned into 12 monthly Parquet files stored locally in `data/raw/`. Spatial reference lookups (`taxi_zone_lookup.csv`) and zone boundary shapefiles (`taxi_zones.shp`) are stored in `data/spatial/`.

3.2 Structural Audit

Using an in-memory DuckDB query engine, we processed all 12 Parquet files in a single unified operation without memory overflow. Table 1 lists the complete schema of the 2025 dataset, comprising 20 columns.

Table 1: NYC Yellow Taxi 2025 Dataset Schema and Variable Categorization

Column Name	Data Type	Category	Description / Role
VendorID	INTEGER	Categorical ID	TPEP provider code
tpep_pickup_datetime	TIMESTAMP	Datetime Variable	Trip start time (Meter engaged)
tpep_dropoff_datetime	TIMESTAMP	Datetime Variable	Trip end time (Meter disengaged)
passenger_count	BIGINT	Numeric	Driver-entered passenger count
trip_distance	DOUBLE	Continuous Numeric	Elapsed trip distance (miles)
RatecodeID	BIGINT	Categorical ID	Final rate code in effect
store_and_fwd_flag	VARCHAR	Flag	Memory buffer transmission flag
PULocationID	INTEGER	Spatial ID	Pickup TLC Taxi Zone ID (1..263)
DOLocationID	INTEGER	Spatial ID	Dropoff TLC Taxi Zone ID (1..263)
payment_type	BIGINT	Categorical	Payment method (1=Credit, 2=Cash)
fare_amount	DOUBLE	Financial Numeric	Metered time-and-distance fare
extra	DOUBLE	Financial Numeric	Miscellaneous extras and surcharges
mta_tax	DOUBLE	Financial Numeric	$0.50 MTA state tax
tip_amount	DOUBLE	Financial Numeric	Credit card tip amount
tolls_amount	DOUBLE	Financial Numeric	Total bridge/tunnel tolls
improvement_surcharge	DOUBLE	Financial Numeric	$0.30 improvement surcharge
total_amount	DOUBLE	Financial Numeric	Total charge to passenger
congestion_surcharge	DOUBLE	Financial Numeric	Standard congestion fee
Airport_fee	DOUBLE	Financial Numeric	Airport pickup fee
cbd_congestion_fee	DOUBLE	Financial Numeric	New 2025 CBD Congestion Pricing Fee

3.3 Spatial Data Structure and Schema Audit

To support spatial aggregations, zone-level demand forecasting, and geographic visualizations, we conducted a structural audit of the spatial reference files stored in `data/spatial/`.

3.4 Taxi Zone Lookup Table (`taxi_zone_lookup.csv`)

The lookup table maps numerical taxi zone identifiers to descriptive geographic attributes, comprising 265 total records and 4 columns, as detailed in Table 2.

Table 2: Taxi Zone Lookup Table Schema (`taxi_zone_lookup.csv`)

Column Name	Data Type	Category	Description / Role
`LocationID`	INTEGER	Primary Identifier	Unique zone identifier (1..265)
`Borough`	VARCHAR	Categorical	NYC Borough name (Manhattan, Queens, etc.)
`Zone`	VARCHAR	Descriptive	Neighborhood or specific zone name
`service_zone`	VARCHAR	Categorical	Service zone classification (Yellow Zone, Boro Zone, etc.)

3.5 Taxi Zones Spatial File (`taxi_zones.shp`)

The spatial vector dataset provides polygon boundaries corresponding to the taxi zones (stored as an ESRI Shapefile bundle with CRS EPSG:2263), comprising 263 distinct polygon records and 7 attributes, as listed in Table 3.

Table 3: Taxi Zones Spatial Vector Schema (`taxi_zones.shp`)

Column Name	Data Type	Category	Description / Role
`OBJECTID`	INTEGER	Identifier	ESRI object feature identifier
`Shape_Leng`	DOUBLE	Numeric Metric	Perimeter length of the zone polygon
`Shape_Area`	DOUBLE	Numeric Metric	Area of the zone polygon
`zone`	VARCHAR	Descriptive	Neighborhood zone name matching lookup table
`LocationID`	INTEGER	Spatial ID / Foreign Key	TLC Taxi Zone ID linking to lookup and trips
`borough`	VARCHAR	Categorical	Borough name classification
`geometry`	GEOMETRY	Spatial Data Type	Polygon / MultiPolygon boundary coordinates (EPSG:2263)

3.6 Table and Spatial Relationships

The trip records link dynamically to the spatial reference files using `PULocationID` and `DOLocationID` as foreign keys that join against the primary key `LocationID` in both the lookup table and the spatial geometry shapefile.

3.7 Data Quality Quantification and Decisions

The audit evaluated $48,722,602$ raw trip records. Table 4 details the findings for missing values, impossible physical measurements, and administrative anomalies, along with the corresponding handling strategy.

Table 4: Data Quality Audit Metrics and Treatment Decisions (Full Year 2025)

Metric / Anomaly Category	Count	% of Total	Initial Treatment Decision
Total Raw Trips	48,722,602	100.00%	Baseline dataset population
Missing Pickup Timestamps	0	0.00%	None required (100% complete)
Missing Dropoff Timestamps	0	0.00%	None required (100% complete)
Unknown Pickup Location (`PULocationID = 264`)	80,873	0.17%	**Filter Out**: Unmapped geographic pickup coordinates
Outside NYC Pickup Location (`PULocationID = 265`)	22,170	0.05%	**Filter Out**: Excluded for internal origin demand modeling
Unknown Dropoff Location (`DOLocationID = 264`)	105,193	0.22%	**Filter Out**: Unmapped geographic dropoff coordinates
Outside NYC Dropoff Location (`DOLocationID = 265`)	219,073	4.49%	**Conditional Retention**: Retained up to 200 miles for out-of-city flow metrics
Missing Distance / Fare Fields	0	0.00%	None required (100% complete)
Invalid Duration ($\leq$ 0 min or $>$ 240 min)	563,024	1.16%	**Filter Out**: Clock reset, meter left running, or cancelled trips
Invalid Trip Distance (Intra $>$ 100mi or Out $>$ 200mi)	2,429	0.005%	**Filter Out**: Extreme spatial anomalies and GPS satellite jumps
Zero Trip Distance ($d = 0$ mi)	1,402,958	2.88%	**Filter Out**: Stationary meter runs / cancellations
Invalid Average Speed ($v > 80$ mph)	560,004	1.15%	**Filter Out**: Telemetry errors and GPS coordinate displacement
Micro-Trip Speed Artifacts ($\Delta t < 1$ min, $v > 70$ mph)	8,543	0.018%	**Filter/Nullify**: Timestamp jitter and GPS rounding glitches
Negative Fare Amount ($f < \$0$)	2,848,620	5.85%	**Filter Out**: Dispute voids, chargebacks, system tests
Invalid Passenger Count ($<$ 0 or $>$ 9 or `null`)	11,611,894	23.83%	**Retain As-Is**: Retained without imputation to prevent data loss.

3.7.1 Rationale for Data Quality Treatment Decisions

To maintain data integrity and prevent distortion in our spatio-temporal demand forecasting models, we established explicit treatment protocols for each identified anomaly category across the $48,722,602$ raw records. The individual rationales and references to representative sample records are structured as follows:

- **Missing Timestamps and Spatial Location IDs (0 records, 0.00%):** *Decision: Retain All. Rationale:* The dataset exhibited complete structural integrity for core operational fields. Because our fundamental unit of analysis is defined by pickup location and timestamp (z, t), having complete coverage in these fields ensures that no records are dropped due to missing spatial keys, eliminating potential sample selection bias at the initial ingestion stage.

- **Invalid Durations (≤ 0 minutes or > 240 minutes, $563,024$ records, 1.16%):** *Decision: Filter Out. Rationale:* A physical taxicab trip must occupy a realistic time interval ($0 < \Delta t \leq 240$ minutes). Records where dropoff precedes pickup or excessive durations exceeding 4 hours occur indicate GPS logging latency, device clock resets, or meters left running indefinitely while parked. As illustrated in Table 5, several records exhibit durations over $1,400$ minutes (~ 24 hours) for short trips under 7 miles, confirming forgotten meters. Retaining these anomalies would severely corrupt vehicle speed calculations and duration-based features.

- **Invalid Trip Distances (Intra-city > 100 mi or Out-of-city > 200 mi, $2,429$ records, 0.005%):** *Decision: Filter Out. Rationale:* While standard intra-city trips are capped at 100 miles to eliminate spatial outliers, legitimate regional long-haul journeys terminating outside NYC (`DOLocationID = 265`) are accommodated with an extended ceiling of 200 miles to preserve authentic interstate economic flows. Records exceeding these conditional thresholds represent severe GPS tracking errors and system telemetry glitches. As shown in Table 6, multiple records log distances exceeding $1,000$ to $4,000$ miles for 10-minute trips, resulting in implausible speeds over $4,000$ mph.

- **Zero-Distance Trips ($d = 0$ miles, $1,402,958$ records, 2.88%):** *Decision: Filter Out. Rationale:* While passengers may occasionally cancel a trip immediately after entering a taxi, zero-distance records predominantly reflect stationary meter testing, administrative diagnostics, or false starts. Table 7 demonstrates that these trips typically last under 3 minutes with identical pickup and dropoff zones while still incurring base fares. Because our spatial aggregation relies on movement across taxi zones, stationary logs distort spatial origin-destination demand densities.

- **Invalid Average Speeds ($v > 80$ mph, $560,004$ records, 1.15%):** *Decision: Filter Out. Rationale:* Municipal yellow cabs operating within urban street grids and regional highways cannot physically or legally sustain average speeds exceeding 80 mph. As documented in Table 11, extreme cases record speeds ranging from 89 mph up to $1,577$ mph over sub-minute intervals

(e.g., traveling 9.20 miles in 21 seconds). These outliers stem from severe GPS satellite coordinate jumps and timestamp desynchronization, which would severely distort velocity profiling and travel-time estimation.

- **Micro-Trip Speed Artifacts ($\Delta t < 1$ min, $v > 70$ mph, $8,543$ records, 0.018%):** *Decision: Filter Out / Conditional Nullification. Rationale:* A subset of short-duration trips ($\Delta t < 1.0$ minute) featuring non-zero distance yields mathematically unstable, exaggerated velocity spikes due to timestamp rounding and high-resolution GPS clock jitter. While standard upper-speed bounds catch many of these, explicitly quantifying these $8,543$ micro-trip artifacts ensures they are handled via conditional kinematic filtering (Option 3)—either nullifying their speed derivatives or filtering them out during the Phase 2 data preparation pipeline to preserve kinematic integrity without sacrificing volume unnecessarily.

- **Negative Fares ($f < \$0$, $2,848,620$ records, 5.85%):** *Decision: Filter Out. Rationale:* Negative fare amounts do not represent legitimate passenger transactions; rather, they correspond to credit card chargebacks, fare dispute reversals, or system administrative adjustments. Table 8 highlights typical negative fare entries (e.g., $-\$3.00$ to $-\$24.00$) for short intra-Manhattan journeys. Including them would severely distort financial summaries and bias regression models exploring fare-demand relationships.

- **Unrecorded or Missing Passenger Counts ($11,611,894$ records, 23.83%):** *Decision: Retain As Is. Rationale:* Approximately 23.83% of raw records exhibited missing (`NULL` / `<NA>`) passenger counts due to driver omission or meter logging habits. As detailed in Table 9, these records represent completely valid commercial trips with normal durations, distances, and fares. Deleting them would cause severe attrition in spatio-temporal trip volume aggregations; hence, they are retained in their native state without artificial imputation.

- **Unknown and Outside NYC Locations (`LocationID = 264, 265`):** *Decision: Filter Out / Conditional Retention. Rationale:* Zone ID 264 designates "Unknown" unmapped spatial areas and is filtered out entirely, as shown in Table 10. Zone ID 265 represents locations outside official TLC zones; trips originating outside NYC (`PULocationID = 265`, $22,170$ records) are excluded to prevent distortion in origin demand modeling, whereas trips terminating outside NYC are retained up to 200 miles to maintain accurate financial and out-commuter flow metrics.

3.7.2 Summary of Cleansing Impact:

Collectively, this rigorous filtering protocol removes $4,707,394$ anomalous records ($\approx 9.66\%$ of the total raw volume), leaving a pristine, high-fidelity baseline population of $44,015,208$ **valid trips** dedicated for downstream spatio-temporal aggregation and hourly demand forecasting.

3.8 Data Flow Architecture

Figure 1 illustrates the end-to-end data pipeline from raw Parquet processing to final spatio-temporal aggregation.

3.9 Empirical Examples of Identified Anomalies

To provide concrete verification of the data quality audit, representative samples from each identified anomaly category are documented in Tables 5–11 below.

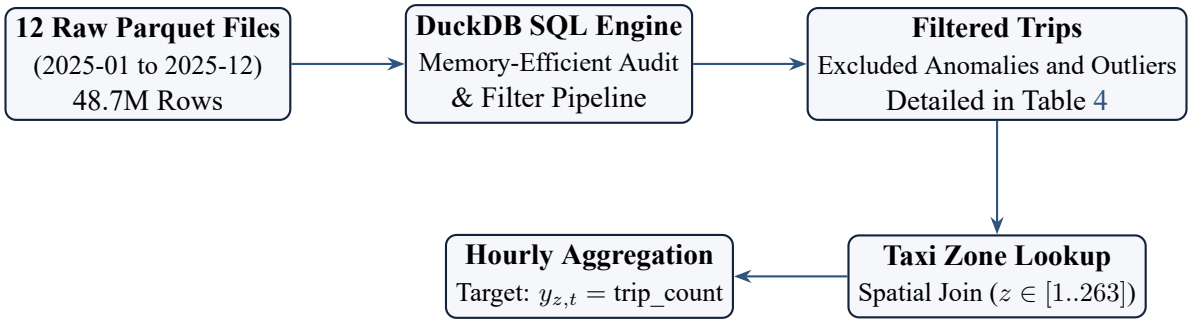


Figure 1: Phase 1 Data Flow and Processing Architecture

Table 5: Sample Records: Invalid or Excessive Duration ($\Delta t > 4$h or ≤ 0 min)

Pickup Time	Dropoff Time	Dur	Dist	Fare	Speed	Pass	PU Zone	DO Zone
2025-01-01 01:26:00	2025-01-02 01:22:08	1436.13	7.06	$37.30	0.29	1	Murray Hill	Boerum Hill
2025-01-01 00:27:08	2025-01-02 00:00:00	1412.87	5.72	$25.40	0.24	1	Kips Bay	Brooklyn Heights
2025-01-01 00:18:22	2025-01-02 00:00:00	1421.63	1.12	$17.00	0.05	1	Clinton West	Clinton East
2025-01-01 02:01:56	2025-01-02 01:17:14	1395.30	0.71	$8.60	0.03	4	West Chelsea/Hudson Yards	East Chelsea
2025-01-01 01:31:10	2025-01-02 01:19:44	1428.57	0.73	$6.50	0.03	1	Union Sq	East Village
2025-01-01 01:48:26	2025-01-02 01:41:57	1433.52	0.99	$8.60	0.04	2	Midtown North	Lincoln Square West
2025-01-01 01:45:26	2025-01-01 11:29:42	584.27	0.56	$6.50	0.06	1	Central Harlem	Queensbridge/Ravenswood
2025-01-01 01:09:24	2025-01-02 01:06:27	1437.05	3.32	$21.20	0.14	1	Upper West Side North	Sutton Place/Turtle Bay North
2025-01-01 01:42:36	2025-01-01 01:42:36	0.00	0.00	$3.00	–	1	Meatpacking/West Village West	Unknown (264)
2025-01-01 00:49:48	2025-01-01 00:49:48	0.00	0.00	$20.06	–	1	Financial District North	Unknown (264)

Table 6: Sample Records: Extreme Distance Anomalies (Intra-city $>$ 100mi or Out-of-city $>$ 200mi)

Pickup Time	Dropoff Time	Dur	Dist	Fare	Speed	Pass	PU Zone	DO Zone
2025-01-09 12:19:06	2025-01-09 13:14:26	55.33	4020.04	$82.80	4359.08	1	JFK Airport	Boerum Hill
2025-01-01 11:32:19	2025-01-01 12:01:14	28.92	122.77	$63.20	254.74	1	Outside of NYC (265)	Erasmus
2025-01-13 17:04:40	2025-01-13 17:15:42	11.03	2001.95	$12.10	10886.74	1	Sutton Place/Turtle Bay North	Lincoln Square West
2025-01-07 23:01:23	2025-01-08 21:36:07	1354.73	265.90	$129.70	11.78	1	Springfield Gardens South	Springfield Gardens South
2025-01-05 18:24:23	2025-01-05 18:42:35	18.20	108.90	$34.50	359.01	2	LaGuardia Airport	Sutton Place/Turtle Bay North
2025-01-08 19:11:56	2025-01-08 19:36:50	24.90	1847.61	$22.60	4452.07	1	Upper East Side South	Union Sq
2025-01-06 08:18:30	2025-01-06 08:33:49	15.32	1472.37	$15.60	5767.72	1	Lenox Hill East	Upper East Side North
2025-01-11 00:00:17	2025-01-11 03:50:52	230.58	224.82	$400.00	58.50	2	Outside of NYC (265)	Outside of NYC (265)
2025-01-11 15:01:08	2025-01-11 15:12:30	11.37	165.91	$17.70	875.77	2	JFK Airport	Springfield Gardens South
2025-01-05 19:42:26	2025-01-05 23:34:00	231.57	206.45	$220.00	53.49	1	Steinway	Outside of NYC (265)

Table 7: Sample Records: Zero-Distance Trips ($d = 0$ miles)

Pickup Time	Dropoff Time	Dur	Dist	Fare	Speed	Pass	PU Zone	DO Zone
2025-01-01 00:43:22	2025-01-01 00:46:18	2.93	0.00	$5.00	0.00	1	Clinton East	Clinton East
2025-01-01 00:24:26	2025-01-01 00:24:34	0.13	0.00	$20.00	0.00	4	Crotona Park East	Crotona Park East
2025-01-01 00:27:40	2025-01-01 00:59:30	31.83	0.00	$50.50	0.00	1	Mott Haven/Port Morris	East New York
2025-01-01 00:14:57	2025-01-01 00:17:44	2.78	0.00	$3.00	0.00	1	East Village	East Village
2025-01-01 00:57:08	2025-01-01 00:57:16	0.13	0.00	$30.00	0.00	3	Lenox Hill West	Lenox Hill West
2025-01-01 00:49:10	2025-01-01 00:49:14	0.07	0.00	$85.00	0.00	4	Lincoln Square West	Lincoln Square West
2025-01-01 00:37:43	2025-01-01 00:37:53	0.17	0.00	$12.00	0.00	1	Lower East Side	Lower East Side
2025-01-01 00:56:49	2025-01-01 00:56:54	0.08	0.00	$20.00	0.00	4	Midtown South	Midtown South
2025-01-01 00:42:42	2025-01-01 00:42:44	0.03	0.00	$3.00	0.00	0	World Trade Center	World Trade Center
2025-01-01 00:49:48	2025-01-01 00:49:48	0.00	0.00	$20.06	–	1	Financial District North	Unknown (264)

Table 8: Sample Records: Negative Fare Amounts ($f < \$0$)

Pickup Time	Dropoff Time	Dur	Dist	Fare	Speed	Pass	PU Zone	DO Zone
2025-01-01 00:11:58	2025-01-01 00:12:31	0.55	0.01	-$3.00	1.09	1	Central Harlem North	Central Harlem North
2025-01-01 00:55:53	2025-01-01 01:06:49	10.93	1.42	-$12.10	7.79	1	East Village	Chinatown
2025-01-01 00:01:41	2025-01-01 00:07:14	5.55	0.71	-$7.20	7.68	1	East Village	Gramercy
2025-01-01 00:29:35	2025-01-01 00:36:02	6.45	0.60	-$7.20	5.58	1	East Village	Lower East Side
2025-01-01 00:50:13	2025-01-01 01:11:20	21.12	3.84	-$24.00	10.91	1	Clinton East	Manhattan Valley
2025-01-01 00:11:44	2025-01-01 00:25:41	13.95	1.88	-$14.20	8.09	2	East Village	Midtown Center
2025-01-01 00:56:12	2025-01-01 01:15:00	18.80	0.97	-$16.30	3.10	1	Midtown Center	Murray Hill
2025-01-01 00:55:54	2025-01-01 01:00:38	4.73	0.69	-$6.50	8.75	1	Kips Bay	UN/Turtle Bay South
2025-01-01 00:51:05	2025-01-01 00:54:12	3.12	0.92	-$6.50	17.71	1	Lincoln Square East	Upper West Side South
2025-01-01 00:09:58	2025-01-01 00:14:28	4.50	0.60	-$6.50	8.00	1	Lenox Hill East	Yorkville West

Table 9: Sample Records: Invalid or Missing Passenger Counts (NULL)

Pickup Time	Dropoff Time	Dur	Dist	Fare	Speed	Pass	PU Zone	DO Zone
2025-01-01 00:39:17	2025-01-01 01:10:42	31.42	6.75	$2.32	12.89	NA	Clinton East	Central Harlem North
2025-01-01 00:33:40	2025-01-01 01:10:34	36.90	3.88	$6.02	6.31	NA	Alphabet City	Clinton East
2025-01-01 00:55:15	2025-01-01 01:43:25	48.17	3.30	$27.55	4.11	NA	Gramercy	Clinton West
2025-01-01 00:41:07	2025-01-01 01:02:39	21.53	2.97	$21.83	8.28	NA	Clinton West	Gramercy
2025-01-01 00:09:32	2025-01-01 00:12:38	3.10	0.00	$22.72	0.00	NA	Meatpacking/West Village West	Meatpacking/West Village West
2025-01-01 00:33:36	2025-01-01 00:54:48	21.20	2.60	$18.42	7.36	NA	Yorkville West	Midtown East
2025-01-01 00:21:42	2025-01-01 00:36:29	14.78	3.55	-$2.35	14.41	NA	Kips Bay	SoHo
2025-01-01 00:56:04	2025-01-01 01:06:11	10.12	1.55	$5.06	9.19	NA	Lenox Hill East	UN/Turtle Bay South
2025-01-01 00:50:25	2025-01-01 00:57:34	7.15	1.04	$14.16	8.73	NA	Lincoln Square East	Upper West Side South
2025-01-01 00:52:42	2025-01-01 01:27:35	34.88	5.45	$23.13	9.37	NA	Little Italy/NoLiTa	Yorkville West

Table 10: Sample Records: Location Anomalies (Zones 264 and 265)

Pickup Time	Dropoff Time	Dur	Dist	Fare	Speed	Pass	PU Zone	DO Zone
2025-01-01 00:52:29	2025-01-01 00:58:00	5.52	0.90	$7.90	9.79	2	Unknown (264)	Lincoln Square East
2025-01-01 00:19:33	2025-01-01 00:30:45	11.20	2.80	$14.20	15.00	1	Unknown (264)	Unknown (264)
2025-01-01 00:32:34	2025-01-01 00:47:43	15.15	1.96	$14.90	7.76	1	Unknown (264)	Unknown (264)
2025-01-01 01:02:16	2025-01-01 01:15:01	12.75	2.17	$14.20	10.21	1	Unknown (264)	Unknown (264)
2025-01-01 00:41:49	2025-01-01 00:49:23	7.57	2.00	$10.70	15.86	5	Unknown (264)	Unknown (264)
2025-01-01 00:59:53	2025-01-01 01:29:44	29.85	5.00	$26.10	10.05	5	Unknown (264)	Unknown (264)
2025-01-01 00:56:15	2025-01-01 01:05:07	8.87	1.40	$10.00	9.47	2	West Village	Unknown (264)
2025-01-01 00:56:06	2025-01-01 01:00:08	4.03	0.40	$5.80	5.95	2	TriBeCa/Civic Center	Unknown (264)
2025-01-01 00:39:55	2025-01-01 00:47:32	7.62	1.60	$10.00	12.60	2	Lincoln Square East	Unknown (264)
2025-01-01 00:49:48	2025-01-01 00:49:48	0.00	0.00	$20.06	–	1	Financial District North	Unknown (264)

Table 11: Sample Records: Invalid Speed Anomalies ($v > 80$ mph)

Pickup Time	Dropoff Time	Dur	Dist	Fare	Speed	Pass	PU Zone	DO Zone
2025-01-01 01:29:00	2025-01-01 01:29:16	0.27	2.20	$54.00	495.00	1	Bushwick South	Bushwick South
2024-12-31 23:51:20	2025-01-01 00:00:00	8.67	12.89	$48.50	89.24	1	LaGuardia Airport	Carroll Gardens
2025-01-01 02:32:52	2025-01-01 02:33:53	1.02	6.10	$30.00	360.00	3	Clinton East	Clinton East
2025-01-01 02:15:12	2025-01-01 02:15:36	0.40	3.40	$12.00	510.00	1	Fort Greene	Fort Greene
2025-01-01 05:43:36	2025-01-01 05:43:57	0.35	9.20	$40.00	1577.14	0	Homecrest	Homecrest
2025-01-01 03:32:51	2025-01-01 03:33:45	0.90	4.00	$3.00	266.67	1	TriBeCa/Civic Center	TriBeCa/Civic Center
2025-01-01 04:09:33	2025-01-01 04:10:06	0.55	1.90	$3.00	207.27	1	Upper West Side South	Upper West Side South
2025-01-01 04:02:52	2025-01-01 04:02:59	0.12	0.80	$3.00	411.43	2	Washington Heights South	Washington Heights South
2025-01-01 01:13:07	2025-01-01 01:13:33	0.43	3.00	$3.00	415.38	1	West Chelsea/Hudson Yards	West Chelsea/Hudson Yards
2025-01-01 04:39:52	2025-01-01 04:39:54	0.03	0.11	$100.00	198.00	1	East Village	Unknown (264)

Table 12: Sample Records: Micro-Trip Speed Artifacts ($\Delta t < 1$ min and $v > 70$ mph)

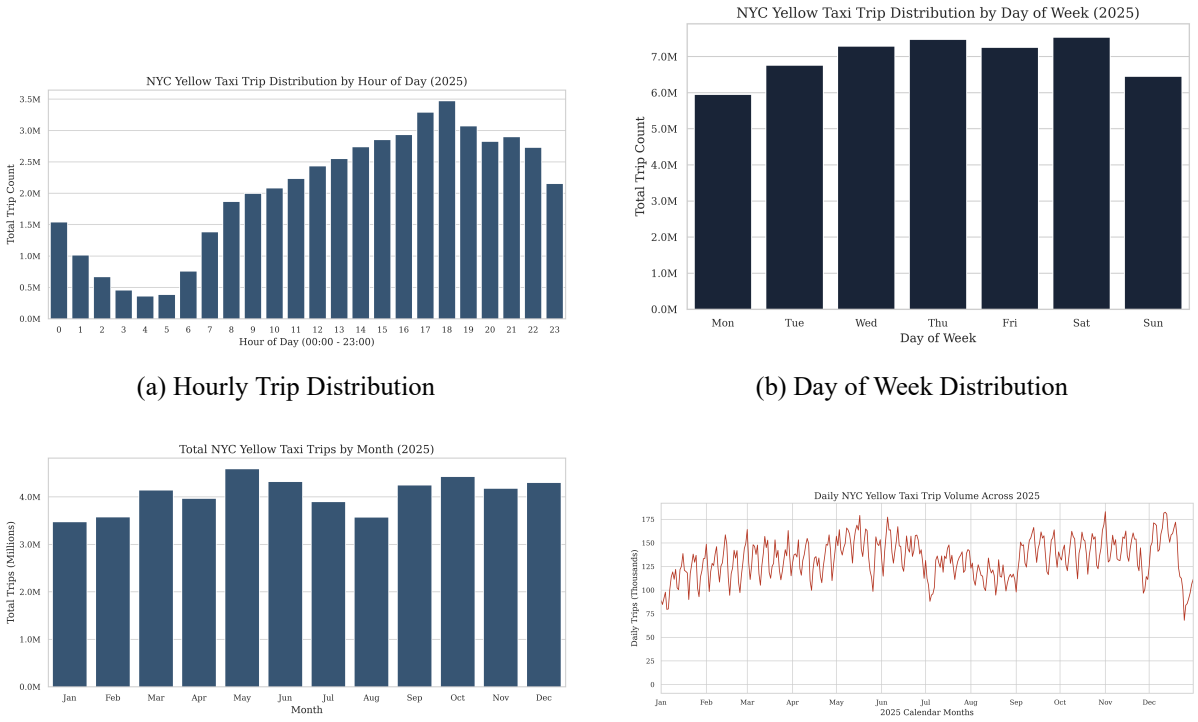
Pickup Time	Dropoff Time	Dur	Dist	Fare	Speed	Pass	PU Zone	DO Zone
2025-01-01 01:29:00	2025-01-01 01:29:16	0.27	2.20	$54.00	495.00	1	Bushwick South	Bushwick South
2025-01-01 05:00:19	2025-01-01 05:00:41	0.37	0.43	$250.00	70.36	2	East Chelsea	East Chelsea
2025-01-01 02:15:12	2025-01-01 02:15:36	0.40	3.40	$12.00	510.00	1	Fort Greene	Fort Greene
2025-01-01 05:43:36	2025-01-01 05:43:57	0.35	9.20	$40.00	1577.14	0	Homecrest	Homecrest
2025-01-01 03:32:51	2025-01-01 03:33:45	0.90	4.00	$3.00	266.67	1	TriBeCa/Civic Center	TriBeCa/Civic Center
2025-01-01 04:09:33	2025-01-01 04:10:06	0.55	1.90	$3.00	207.27	1	Upper West Side South	Upper West Side South
2025-01-01 04:02:52	2025-01-01 04:02:59	0.12	0.80	$3.00	411.43	2	Washington Heights South	Washington Heights South
2025-01-01 01:13:07	2025-01-01 01:13:33	0.43	3.00	$3.00	415.38	1	West Chelsea/Hudson Yards	West Chelsea/Hudson Yards
2025-01-01 04:55:51	2025-01-01 04:55:53	0.03	2.02	$20.00	3636.00	2	Unknown (264)	Unknown (264)
2025-01-01 04:39:52	2025-01-01 04:39:54	0.03	0.11	$100.00	198.00	1	East Village	Unknown (264)

4 Exploratory Data Analysis (EDA)

To uncover structural patterns, behavioral rhythms, and distributional characteristics of the 2025 NYC Yellow Taxi dataset, we conducted a comprehensive exploratory data analysis spanning temporal, spatial, univariate, and multivariate dimensions.

4.1 Temporal Distributions and Time Audit

Passenger demand in urban environments exhibits pronounced multi-scale temporal non-stationarity. Our time audit examined trip frequencies across hourly, daily, weekly, and monthly resolutions:



(a) Hourly Trip Distribution

(b) Day of Week Distribution

(c) Monthly Trip Volume

(d) Daily Time Series

Figure 2: Temporal Distribution and Time Series Analysis of NYC Yellow Taxi Trips (2025)

- **Intra-Day Hourly Patterns:** Trip volumes follow a distinct bimodal commuter curve (Figure 2a). Activity hits its nightly trough between 03:00 and 05:00 AM ($\approx$ 0.35M trips), ramps up steadily through the morning, and peaks sharply during the evening rush hour at 18:00 (6:00 PM) with nearly 3.5M annual originations.
- **Inter-Day Weekly Patterns:**

  Weekly activity (Figure 2b) remains robust from Tuesday through Saturday, with Saturday recording the highest aggregate volume ($\approx$ 7.5M trips), driven by weekend entertainment and nightlife mobility. Monday experiences a relative weekly lull ($\approx$ 5.9M trips).
- **Monthly & Longitudinal Trends:** Monthly aggregation (Figure 2c) highlights peak seasonal volume in May ($\approx$ 4.6M trips), while the daily time-series trend across all 365 calendar days of 2025 (Figure 2d) confirms complete temporal data continuity with zero missing daily partitions and consistent weekly oscillations.

4.2 Exploratory Data Analysis: Trip Distance

The `trip_distance` variable records the total elapsed distance of a journey in miles as measured by the taximeter. Analyzing the distribution of trip distance before and after cleaning provides critical insight into how physical anomalies distort summary metrics and demonstrates the efficacy of our preprocessing protocols.

Table 13: Pre- and Post-Cleansing Descriptive Statistics for Trip Distance (2025 Dataset)

Metric	**Pre-Cleansing (Raw)**	**Post-Cleansing (Clean)**
Total Records (N)	48,722,602	44,015,208
Mean (μ)	6.84 miles	3.49 miles
25th Percentile (Q_1)	1.04 miles	1.10 miles
Median (Q_2)	1.85 miles	1.89 miles
75th Percentile (Q_3)	3.70 miles	3.73 miles
Standard Deviation (σ)	649.36 miles	4.32 miles
Interquartile Range (IQR)	2.66 miles	2.63 miles
Skewness	240.90	3.09

4.2.1 Statistical Characteristics and Cleansing Impact Analysis

As detailed in Table 13, the raw dataset exhibits extreme sensitivity to telemetry noise and recording corruptions:

- **Impact on Dispersion and Variance:** In the pre-cleansing dataset, the standard deviation ($\sigma = 649.36$ miles) was nearly $100\times$ larger than the mean, driven by extreme GPS satellite jump errors logging distances in excess of $4,000$ miles for sub-hour trips. Following our cleaning pipeline, the standard deviation stabilizes at 4.32 miles, accurately capturing natural variability across local street hails and regional trips.
- **Impact on Skewness:** Raw telemetry errors introduced an immense positive skewness of 240.90. Removing non-physical artifacts reduced skewness by over 98% down to a natural right-skewed value of 3.09.
- **Stability of Robust Non-Parametric Measures:** While the mean dropped sharply from 6.84 miles to 3.49 miles (due to the removal of multi-thousand-mile GPS glitches and zero-distance stationary records), robust order statistics remained remarkably stable. The median shifted slightly from 1.85 to 1.89 miles, and the IQR moved from 2.66 to 2.63 miles, proving that our cleansing protocol removes gross artifacts without corrupting the core underlying trip distribution.

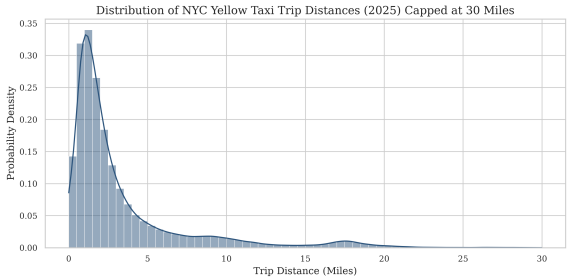
The probability density function (Figure 3a) shows a dense concentration of mass below 3 miles, representing routine intra-Manhattan journeys, followed by an extended positive tail for airport and regional transfers.

4.2.2 Long-Distance Trips and Outlier Analysis (>30 Miles)

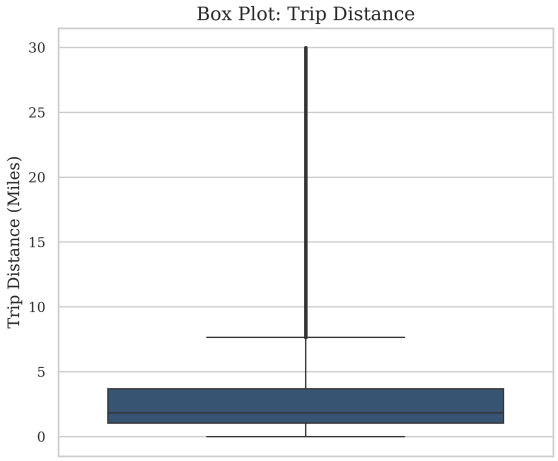
A detailed query of clean trips exceeding 30 miles yields a total of **40,395 valid long-distance trips** across the 2025 dataset ($\approx 0.092\%$ of the pristine baseline population). Table 14 lists a representative sample of these extreme-distance records ordered by descending distance.

4.2.3 Justification for Retaining Outliers Beyond the 1.5×IQR Rule

Applying Tukey's standard fence rule ($Q_3 + 1.5 \times \text{IQR}$) yields an outlier threshold of $3.73 + 1.5 \times (2.63) \approx 7.675$ miles. While standard statistical pipelines might trim records beyond 7.675 miles, we retain them based on the following justifications:



(a) Probability Density Distribution Capped at 30 Miles



(b) Box Plot (Capped at 30 Miles)

Figure 3: Distribution and Box Plot Analysis for Filtered Trip Distance (2025)

Table 14: Sample of Long-Distance Yellow Taxi Trips Exceeding 30 Miles (2025)

Pickup Time	Dropoff Time	Dur (min)	Dist (mi)	Fare ($)	Speed (mph)	Pass.	Pickup Zone	Dropoff Zone
2025-08-30 19:16	2025-08-30 22:24	188.35	197.54	$450.00	62.93	1	Flushing Meadows-Corona Park	Outside of NYC
2025-05-22 01:43	2025-05-22 05:23	220.20	197.27	$550.00	53.75	1	JFK Airport	Outside of NYC
2025-03-13 21:33	2025-03-14 00:51	197.85	195.72	$450.00	59.35	1	JFK Airport	Outside of NYC
2025-12-31 16:32	2025-12-31 20:05	213.77	195.33	$1295.90	54.83	2	JFK Airport	Outside of NYC
2025-04-18 15:34	2025-04-18 19:08	213.95	194.85	$1303.60	54.64	1	JFK Airport	Outside of NYC
2025-11-16 11:13	2025-11-16 14:49	216.82	192.97	$460.23	53.40	NA	World Trade Center	Outside of NYC
2025-06-06 10:00	2025-06-06 13:46	225.58	191.81	$800.00	51.02	3	East Chelsea	Outside of NYC
2025-03-02 06:12	2025-03-02 09:32	200.53	190.50	$650.00	57.00	1	Springfield Gardens South	Outside of NYC
2025-01-21 05:17	2025-01-21 08:19	181.80	188.88	$1309.20	62.34	1	West Concourse	Outside of NYC
2025-01-18 21:07	2025-01-19 00:20	192.55	183.95	$503.59	57.32	NA	Clinton East	Outside of NYC

1. **Legitimate Out-of-Jurisdiction and Regional Transit Activity:** As evidenced in Table 14, extreme distance records originate at major transit nodes (JFK, World Trade Center) and terminate at `Outside of NYC` destinations. These $40,395$ trips represent real, high-value commercial journeys.

2. **Economic Integrity and Revenue Preservation:** Removing trips exceeding 7.675 miles would eliminate a major share of total revenue. Out-of-city journeys command premium fares (up to $1,309.20 in our sample) and contribute significantly to overall fleet earnings and congestion fees.

3. **Prevention of Truncation Bias in Spatial Aggregation:** Trimming valid long-distance journeys would distort origin-destination flow matrices and underestimate regional vehicle turnover times.

4. **Methodological Accommodation over Trimming:** Right-skewness (Skewness $= 3.09$) is accommodated using non-parametric summary measures (median $= 1.89$ miles, IQR $= 2.63$ miles) and axis capping in visualizations (e.g., 30 miles in distribution plots) to maintain graphical readability while preserving 100% of authentic economic activity.

4.3 Exploratory Data Analysis: Fare Amount

The `fare_amount` variable represents the time-and-distance fare in USD calculated by the taximeter at the conclusion of each trip. Comparing the statistical properties of the fare amount before and after data cleansing illustrates the structural impact of removing negative fare anomalies.

Table 15: Pre- and Post-Cleansing Descriptive Statistics for Fare Amount (2025 Dataset)

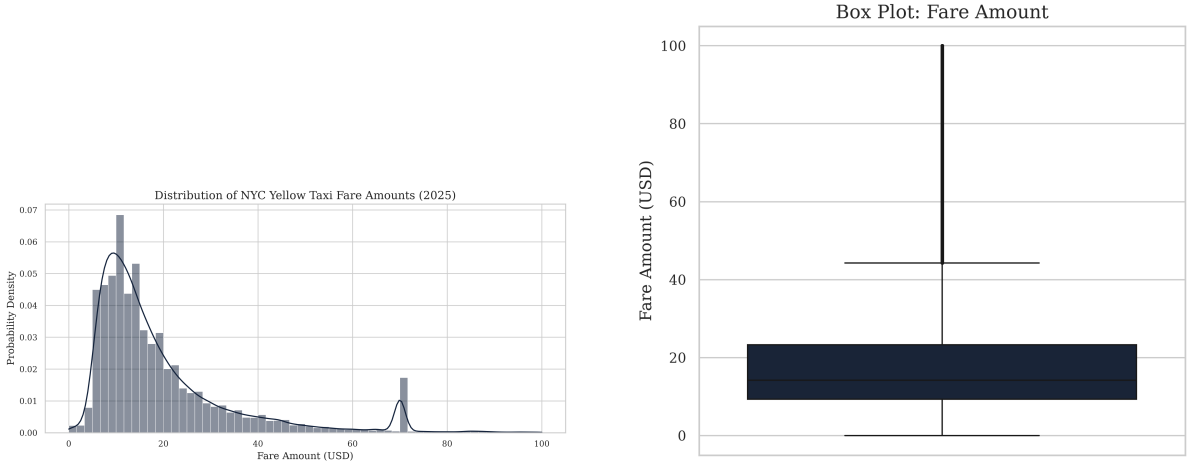
Metric	**Pre-Cleansing (Raw)**	**Post-Cleansing (Clean)**
Total Records (N)	48,722,602	44,015,208
Mean (μ)	\$18.42	\$20.04
25th Percentile (Q_1)	\$8.60	\$9.30
Median (Q_2)	\$13.58	\$14.20
75th Percentile (Q_3)	\$22.60	\$23.30
Standard Deviation (σ)	\$142.85	\$149.75
Interquartile Range (IQR)	\$14.00	\$14.00
Skewness	5,030.24	4,832.47

4.3.1 Statistical Characteristics and Cleansing Impact Analysis

As summarized in Table 15, the median fare increases slightly from $\$13.58$ in the raw dataset to $\$14.20$ after cleaning, while the mean increases from $\$18.42$ to $\$20.04$.

- **Rationale for Mean Increase:** The upward shift in the mean fare is a direct consequence of filtering out the $2,848,620$ negative fare records (credit card chargebacks, dispute reversals, and administrative adjustments) present in the raw data. Removing these sub-zero financial entries eliminates downward distortion, thereby reflecting the true positive economic value of commercial passenger trips.
- **Distributional Shape:** The distribution continues to exhibit extreme positive skewness ($4,832.47$), driven by a heavy concentration of short, low-cost urban trips combined with an extended right-hand tail of high-value journeys.

The probability density function (Figure 4a) reveals a prominent unimodal peak representing standard local city rides ($\$10 - \$15$), alongside a distinct structural cluster or secondary bump around $\$70$. This secondary surge corresponds directly to regulated flat-rate airport trips (such as historical or standard JFK airport routes), highlighting a multimodal element embedded within the financial structure of the fleet.



(a) Probability Density Distribution

(b) Box Plot (Capped at \$100)

Figure 4: Distribution and Box Plot Analysis for Fare Amount (2025)

4.3.2 Justification for Retaining Outliers Beyond the 1.5×IQR Rule

Under standard Tukey fence criteria, the upper bound for outliers is defined as $Q_3 + 1.5 \times \text{IQR}$. Given an approximate third quartile $Q_3 = \$23.30$ and an IQR of $\$14.00$, this traditional threshold flags any fare exceeding approximately $\$44.30$ as an outlier.

However, **these high-value fare observations are strictly retained rather than trimmed** for the following operational and economic reasons:

1. **Commercial Validity of Airport Flat Rates:** Fares exceeding the $\$44.30$ threshold are not data errors or anomalies; they represent authorized high-value commercial transactions, most notably the regulated flat rates for trips between Manhattan and major regional airports ($\approx \$70$, clearly visible in Figure 4a). Deleting these records would discard crucial airport demand data.

2. **Prevention of Economic Distortion:** Trimming observations beyond the $1.5 \times$ IQR fence would artificially deflate total revenue estimates, average ticket sizes, and macro-economic metrics across the dataset. Retaining them preserves the true economic distribution of NYC yellow taxi operations.

3. **Robust Analytical Presentation:** Rather than destructive data filtering, extreme financial values are managed through robust location-and-spread metrics (median and IQR) and visual capping ($\$100$ ceiling in Figure 4b) to ensure graphical readability while maintaining complete data fidelity.

4.4 Exploratory Data Analysis: Passenger Count

The `passenger_count` variable tracks the number of passengers reported by the driver during the trip. Across the 2025 dataset, while unrecorded or missing passenger entries account for $11,611,894$ records (23.83% of total raw volume), the subset of valid, explicitly recorded trips comprises $35,174,850$ records. Passenger occupancy patterns across this valid subset demonstrate the heavy prevalence of single-occupancy transit typical of urban yellow taxi services.

Table 16: Descriptive Statistics for Valid Passenger Count (2025 Dataset)

Metric	Value
Total Valid Recorded Trips	35,174,522
Mode	1
Mean (μ)	1.29
25th Percentile (Q_1)	1.00
Median (Q_2)	1.00
75th Percentile (Q_3)	1.00
Standard Deviation (σ)	0.72
Interquartile Range (IQR)	0.00
Skewness	2.97

4.4.1 Statistical Characteristics and Distribution Shape

As summarized in Table 16, for the subset of explicitly recorded trips, the mode, 25th percentile (Q_1), median (Q_2), and 75th percentile (Q_3) are all strictly 1.00, with a mean (μ) of 1.29 and an IQR of 0.00 due to the overwhelming concentration of single-passenger rides. The distribution exhibits a positive skewness of 2.97.

As visualized in the frequency distribution (Figure 5), which excludes the $11,611,894$ unrecorded or missing entries to ensure clean graphical representation, passenger counts follow a discrete categorical distribution where:

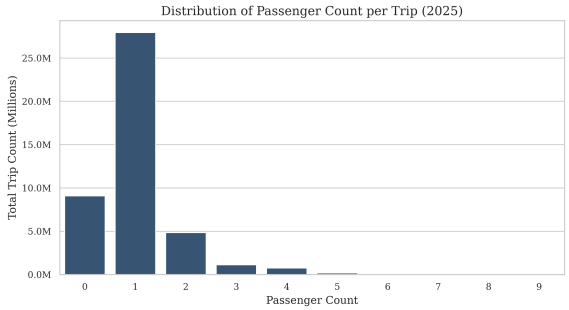


Figure 5: Trip Count by Passenger Occupancy (Excluding Unrecorded Values)

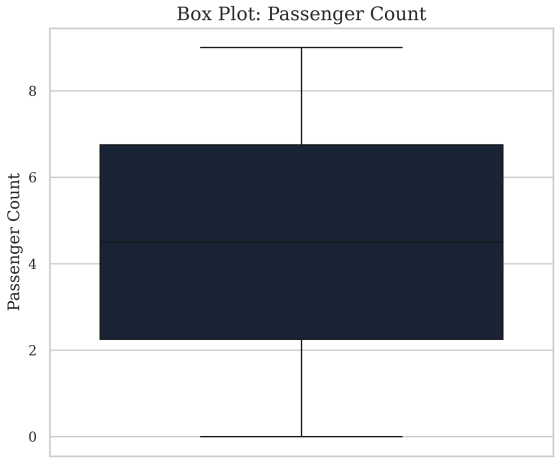


Figure 6: Box Plot for Discrete Occupancy Tiers

Figure 7: Distribution and Box Plot Analysis for Passenger Count (2025)

1. **Single Occupancy (1 Passenger):** Dominates fleet utilization, representing the vast majority of recorded trips.
2. **Solo/Zero-Passenger Records (0 Passengers):** Account for a secondary cluster of trips, typically representing administrative entries, driver-only transits, or unassigned meter logs permitted under TLC guidelines.
3. **Multi-Passenger Tiers:** Frequencies drop off sharply for groups of 2, 3, 4, and 5+ passengers, reflecting physical vehicle capacity limits and typical party sizes for standard yellow cabs.

4.4.2 Methodological Note on Discrete Categorical Variables and Graphical Exclusions

Because `passenger_count` is an integer-based discrete variable rather than a continuous metric, traditional outlier bounds like Tukey's $1.5 \times$ IQR rule do not apply in the standard continuous sense. Furthermore, unrecorded and missing passenger count records (23.83% of the raw dataset) are explicitly excluded from both the frequency distribution plot (Figure 5) and the box plot (Figure 6) to prevent null or undefined categories from distorting the visual analysis of active passenger occupancy. Retaining these records in the broader data pipeline while isolating them from univariate graphical rendering ensures both comprehensive data retention and precise visual interpretation.

4.5 Exploratory Data Analysis: Tip Amount

The `tip_amount` variable records the gratuity provided by the passenger, automatically captured for electronic payments and optionally logged for cash transactions. Comparing the statistical properties of the tip amount before and after data cleansing illustrates the structural impact of removing gross anomalies.

4.5.1 Statistical Characteristics and Cleansing Impact Analysis

As summarized in Table 17, the median tip shifts from \$2.00 in the raw dataset to \$2.39 after cleaning, while the mean increases from \$2.84 to \$3.04.

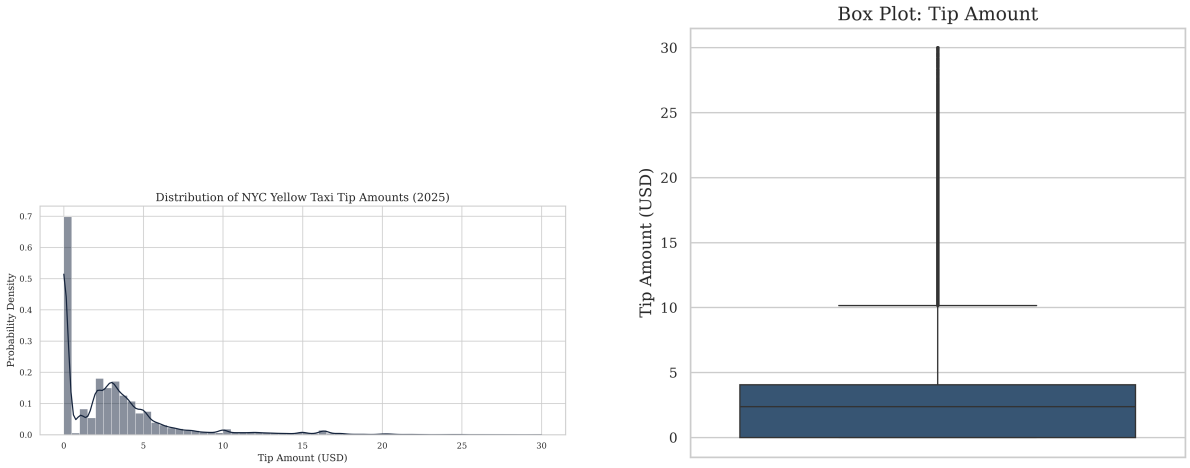
- **Impact of Cleansing:** The upward adjustment in both median and mean gratuities stems from filtering out unmapped or erroneous records (such as negative-fare dispute reversals and zero-distance system tests) that corrupted financial reporting. Removing these distortions yields a cleaner representation of true passenger tipping behavior.

Table 17: Pre- and Post-Cleansing Descriptive Statistics for Tip Amount (2025 Dataset)

Metric	**Pre-Cleansing (Raw)**	**Post-Cleansing (Clean)**
Total Records (N)	48,722,602	44,015,208
Mean (μ)	\$2.84	\$3.04
25th Percentile (Q_1)	\$0.00	\$0.00
Median (Q_2)	\$2.00	\$2.39
75th Percentile (Q_3)	\$3.99	\$4.07
Standard Deviation (σ)	\$3.97	\$3.96
Interquartile Range (IQR)	\$3.99	\$4.07
Skewness	5.28	4.53

- **Distributional Shape:** The distribution continues to exhibit strong positive skewness (4.53) and significant dispersion ($\sigma = \$3.96$).

The probability density function (Figure 8a) reveals a distinct zero-inflated structure, characterized by a massive spike at $\$0.00$. This prominent zero-inflation corresponds primarily to cash-paid trips (where gratuities are handed directly to drivers and unrecorded digitally) as well as zero-tip credit card transactions. Beyond the zero origin, the distribution forms a smooth right-skewed hump peaking between $\$2.00$ and $\$3.00$, accompanied by minor multimodal bumps at round currency increments (such as $\$10.00$ and $\$15.00$) representing fixed or percentage-based gratuities on longer journeys.



(a) Probability Density Distribution

(b) Box Plot (Capped at \$30)

Figure 8: Distribution and Box Plot Analysis for Tip Amount (2025)

4.5.2 Justification for Retaining Outliers Beyond the 1.5×IQR Rule

Under standard Tukey fence criteria, the upper bound for outliers is defined as $Q_3 + 1.5 \times \text{IQR}$. Given a third quartile $Q_3 = \$4.07$ and an IQR of $\$4.07$, this traditional threshold flags any tip exceeding approximately $\$10.175$ as an outlier candidate for removal.

However, **these high-value tip observations are strictly retained rather than trimmed** for the following operational and economic reasons:

1. **Commercial Validity of Generous Gratuities:** Tips exceeding $\$10.175$—and extending up to the plotted $\$30.00$ ceiling (Figure 8b)—are authentic economic transactions. They typically accompany high-fare airport transfers, long-distance inter-borough trips, or exceptional customer service satisfaction.

2. **Prevention of Economic Distortion:** Trimming observations beyond the $1.5 \times$ IQR fence would artificially depress driver earnings estimations, distort cashless economic modeling, and bias analyses exploring the relationship between fare size and tip generosity.

3. **Robust Analytical Presentation:** Rather than destructive data deletion, extreme gratuity values are accommodated through robust summary statistics (median and IQR) and visual capping ($30 ceiling in Figure 8b) to ensure graphical clarity without sacrificing dataset integrity.

4.6 Exploratory Data Analysis: Trip Duration

The `trip_duration_minutes` variable measures the elapsed travel time in minutes, computed as the exact difference between drop-off and pick-up timestamps (Δt). Comparing the statistical properties and visual distributions of trip duration before and after data cleansing illustrates the structural impact of removing extreme telemetry and clock-reset anomalies.

Table 18: Pre- and Post-Cleansing Descriptive Statistics for Trip Duration (2025 Dataset)

Metric	Pre-Cleansing (Raw)	Post-Cleansing (Clean)
Total Records (N)	48,722,602	44,015,208
Mean (μ)	17.36 minutes	17.23 minutes
25th Percentile (Q_1)	8.12 minutes	8.30 minutes
Median (Q_2)	13.48 minutes	13.53 minutes
75th Percentile (Q_3)	21.48 minutes	21.52 minutes
Standard Deviation (σ)	28.29 minutes	13.82 minutes
Interquartile Range (IQR)	13.37 minutes	13.22 minutes
Skewness	-69.38	2.49

4.6.1 Statistical Characteristics and Cleansing Impact Analysis

As summarized in Table 18, data cleansing significantly stabilized the dataset's dispersion and distributional shape:

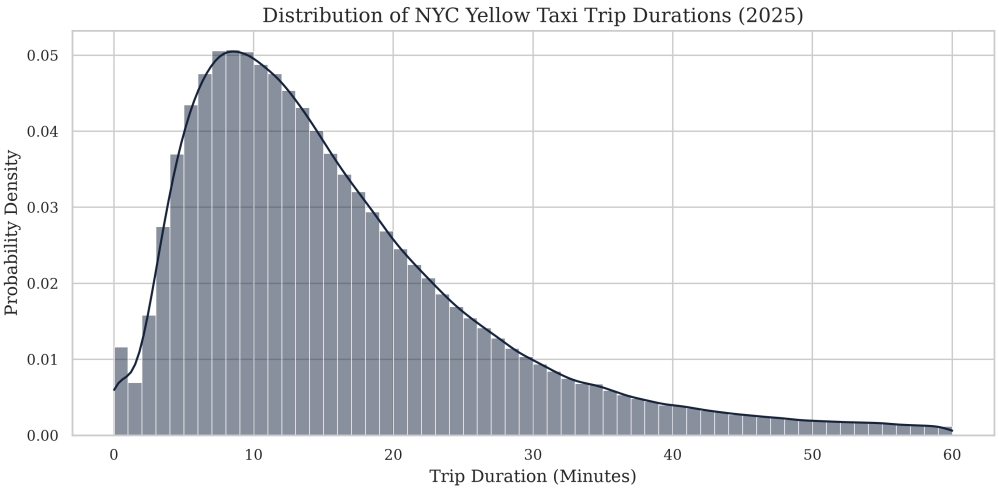
- **Impact on Dispersion and Variance:** In the raw dataset, extreme clock-reset errors and meters left running for 24+ hours inflated the standard deviation to 28.29 minutes. Filtering out durations exceeding 240 minutes and non-positive intervals reduced the standard deviation by over 50% down to 13.82 minutes.

- **Correction of Skewness:** The raw dataset exhibited a heavily distorted negative skewness (-69.38) driven by timestamp anomalies. Removing these errors restored a natural, right-skewed distribution (Skewness $= 2.49$), reflecting an operational structure dominated by short urban movements coupled with an extended tail of prolonged cross-borough or congested journeys.

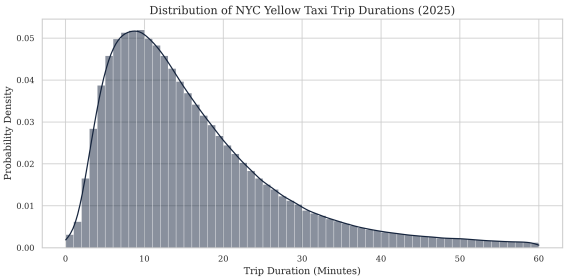
Visual inspection further highlights this transformation. Figure 9a displays the raw pre-cleansing distribution, where distortion artifacts and extreme tails obscure core operational patterns. In contrast, the filtered probability density function (Figure 9b) and box plot (Figure 9c) display a smooth, unimodal distribution rising steeply from zero, peaking sharply between 8 and 12 minutes (representing standard intra-neighborhood and mid-Manhattan trips), and tapering into a natural right-hand tail corresponding to extended regional commutes.

4.6.2 Justification for Retaining Outliers Beyond the 1.5×IQR Rule

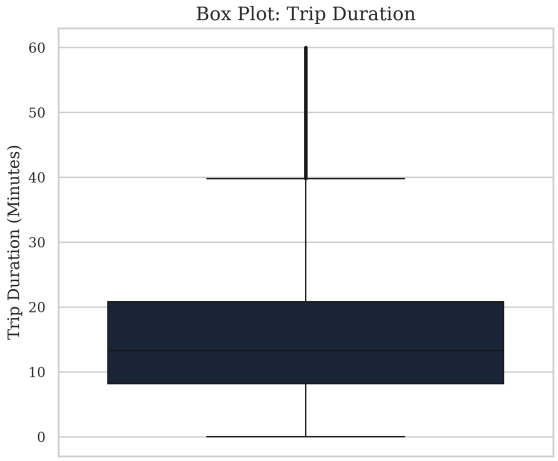
Under standard Tukey fence criteria, the upper bound for outliers is defined as $Q_3 + 1.5 \times \text{IQR}$. Given a third quartile $Q_3 = 21.52$ minutes and an IQR of 13.22 minutes, this traditional threshold flags any trip lasting longer than approximately 41.35 minutes as an outlier candidate for removal.



(a) Pre-Cleansing Trip Duration Distribution (Showing Raw Anomalies)



(b) Post-Cleansing Probability Density



(c) Box Plot (Capped at 60 Minutes)

Figure 9: Pre- and Post-Cleansing Distribution and Box Plot Analysis for Trip Duration (2025)

However, **these high-value duration observations are strictly retained rather than trimmed** for the following operational and modeling reasons:

1. **Legitimate Traffic and Spatial Realities:** Trips exceeding 41.35 minutes—and extending up to the plotted 60-minute ceiling (Figure 9c)—are authentic transportation events. They capture severe peak-hour traffic congestion bottlenecks, cross-borough transits (e.g., Brooklyn to Manhattan), and long-distance airport runs (e.g., Lower Manhattan to JFK or Newark airports).

2. **Prevention of Spatio-Temporal Distortion:** Trimming records beyond the $1.5 \times$IQR fence would artificially truncate travel times, biasing average speed calculations, distorting peak congestion modeling, and eliminating critical rush-hour demand profiles.

3. **Robust Analytical Presentation:** Rather than destructive data filtering, extreme duration values are accommodated through robust summary statistics (median and IQR) and visual capping (60-minute ceiling in Figure 9c) to ensure graphical clarity without compromising dataset integrity.

4.7 Exploratory Data Analysis: Average Trip Speed

The `speed_mph` variable captures the effective operational velocity of each journey in miles per hour, derived as the ratio of trip distance to elapsed travel duration ($v = \frac{d}{\Delta t} \times 60$). Comparing the statistical properties of average speed before and after data cleansing illustrates the structural impact of removing extreme telemetry and GPS satellite jump errors.

Table 19: Pre- and Post-Cleansing Descriptive Statistics for Average Trip Speed (2025 Dataset)

Metric	**Pre-Cleansing (Raw)**	**Post-Cleansing (Clean)**
Total Records (N)	46,775,191	44,015,208
Mean (μ)	11.06 mph	11.00 mph
25th Percentile (Q_1)	7.01 mph	6.99 mph
Median (Q_2)	9.48 mph	9.43 mph
75th Percentile (Q_3)	13.14 mph	13.04 mph
Standard Deviation (σ)	6.40 mph	6.34 mph
Interquartile Range (IQR)	6.13 mph	6.05 mph
Skewness	1.83	1.85

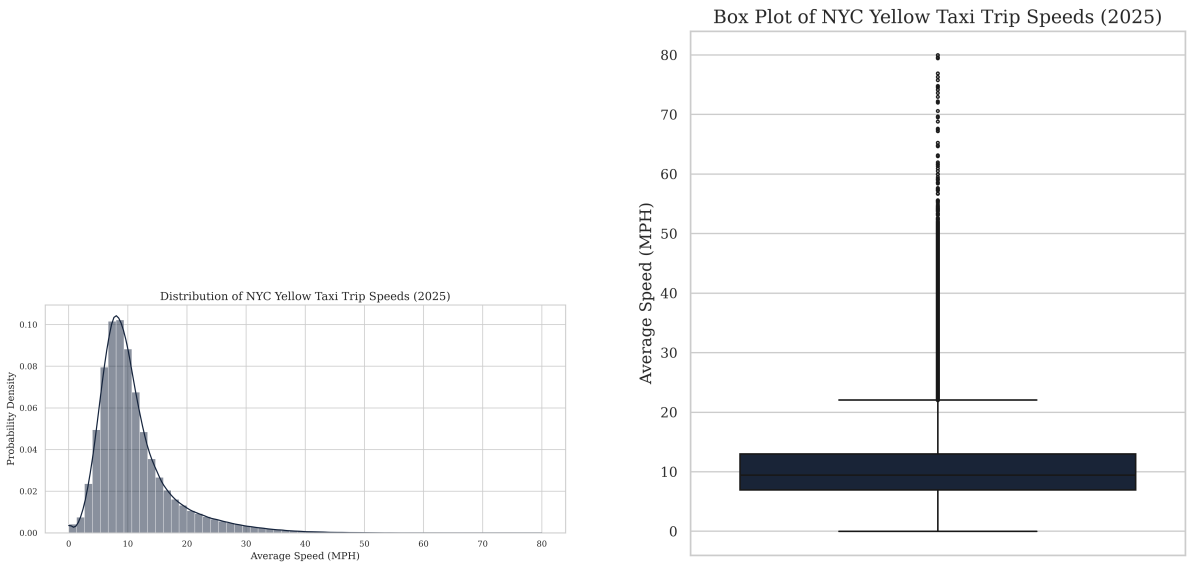
4.7.1 Statistical Characteristics and Cleansing Impact Analysis

As detailed in Table 19, data cleansing successfully removes physically impossible velocity spikes (ranging from 80 mph up to over $1,000$ mph due to timestamp desynchronization) while leaving core urban operational metrics remarkably stable:

- **Impact of Cleansing:** The median speed shifts slightly from 9.48 mph in the raw dataset to 9.43 mph after cleaning, while the mean adjusts from 11.06 mph to 11.00 mph. Removing extreme telemetry outliers prevents artifact-driven inflation, yielding a precise reflection of urban transit velocity.

- **Distributional Shape:** The positive skewness (1.85) and moderate dispersion ($\sigma = 6.34$ mph) reflect the dense, stop-and-go nature of street-hail taxi operations in New York City.

The probability density function (Figure 10a) demonstrates a unimodal, right-skewed distribution with a sharp modal peak concentrated between 7 and 10 mph. This dominant concentration corresponds to intra-Manhattan street grid navigation, characterized by frequent traffic signal delays, pedestrian crossings, and severe urban congestion. The extended right-hand tail represents high-speed arterial corridors, such as highway transits along the FDR Drive, West Side Highway, and expressway connections to regional airports.



(a) Probability Density Distribution

(b) Box Plot (Capped at 80 MPH)

Figure 10: Distribution and Box Plot Analysis for Average Trip Speed (2025)

4.7.2 Justification for Retaining Speeds Beyond the 1.5×IQR Rule

Under standard Tukey fence criteria, observations exceeding $Q_3 + 1.5 \times \text{IQR}$ are classified as upper-bound outliers. Given a third quartile $Q_3 = 13.04$ mph and an IQR of 6.05 mph, this traditional formula flags any average speed exceeding approximately 22.11 mph as an outlier candidate for deletion.

However, **these higher-speed observations are strictly retained up to the physical filtering threshold (**80 **mph)** for the following operational and empirical reasons:

1. **High-Speed Arterial and Highway Realities:** Average speeds between 22 mph and 65 mph (as depicted in Figure 10b) are authentic, legally permitted transportation events. They primarily occur on expressways, parkways, and bridges during late-night hours or on dedicated airport corridors (e.g., Van Wyck Expressway to JFK Airport or Grand Central Parkway to LaGuardia Airport).

2. **Prevention of Systematic Commute Erasure:** Eliminating records above 22.11 mph would systematically erase off-peak and outer-borough airport transits, artificially deflating overall fleet velocity estimates and biasing spatio-temporal traffic models.

3. **Telemetry Filtering Threshold (**80 **mph):** To prevent severe GPS satellite jumps and clock anomalies from corrupting the analytical pipeline, an upper boundary is enforced at 80 mph ($v \leq 80$ mph). This retains 100% of legitimate transportation activity while removing physically impossible telematics errors.

5 Spatial Exploratory Data Analysis and Geographic Distribution

To understand the spatial dynamics and geographic concentration of the 2025 NYC Yellow Taxi dataset, trips were mapped and aggregated across official NYC TLC taxi zones and boroughs. Due to high-density commercial hubs (such as Midtown Manhattan and major transport terminals), standard linear visualizations often compress outer-borough variations. Therefore, both linear and logarithmic spatial density analyses were conducted alongside borough-level trip volume breakdowns.

5.1 NYC Taxi Zones and Borough Breakdown

The study area encompasses all official NYC TLC taxi zones categorized across the five main boroughs (Manhattan, Brooklyn, Queens, the Bronx, and Staten Island), alongside Newark Liberty International Airport (EWR). As illustrated in Figure 11, Manhattan represents a high-density, highly fragmented zone structure, whereas outer boroughs such as Queens and Staten Island feature larger geographic zones with distinct travel patterns.

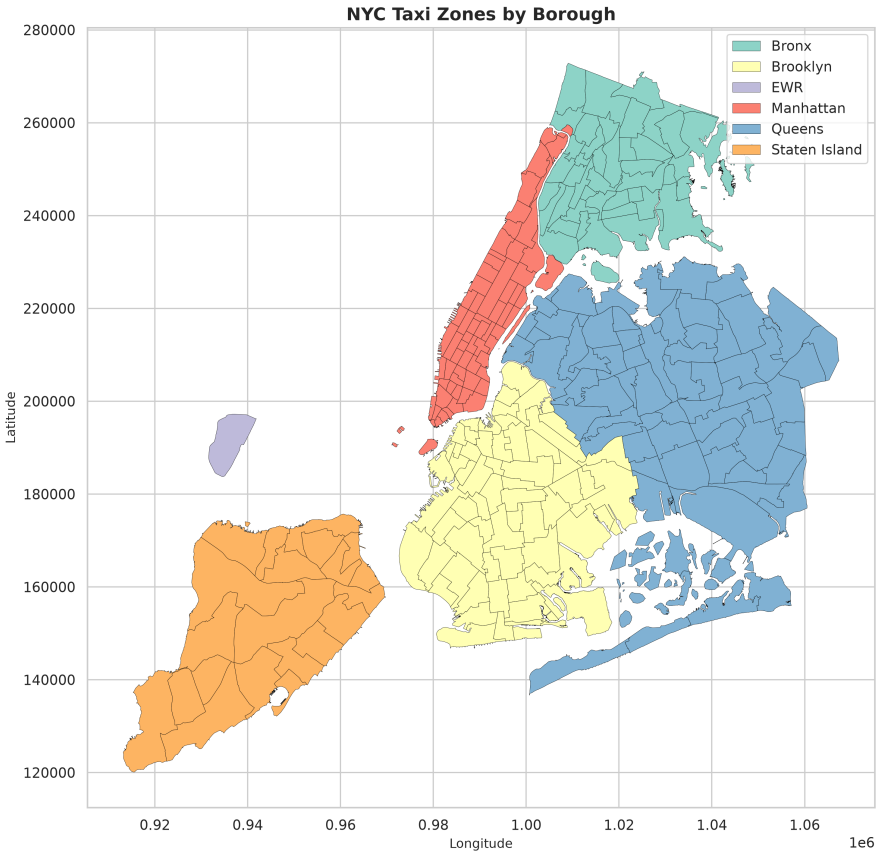


Figure 11: NYC Taxi Zones categorized by Borough.

5.2 Borough-Level Pickup and Drop-off Volume Distribution

Aggregating trip activity by borough reveals a pronounced spatial imbalance between Manhattan and the rest of New York City. Figure 12 compares the volume of pickups and drop-offs across all six administrative jurisdictions.

Key observations from the borough trip distribution include:

- **Manhattan Dominance:** Manhattan accounts for the overwhelming majority of both total pickups and drop-offs, reflecting its status as the city's primary commercial and tourism epicenter.
- **Asymmetric Outer-Borough Trips:** Queens and Brooklyn record substantial drop-off volumes relative to their pickup counts, indicating a high prevalence of inbound commuter or late-night trips returning from Manhattan.
- **Sparser Peripheral Demand:** The Bronx, Staten Island, and EWR register minimal overall Yellow Taxi volume, pointing to localized demand patterns largely served by alternative transport modes or local taxi services.

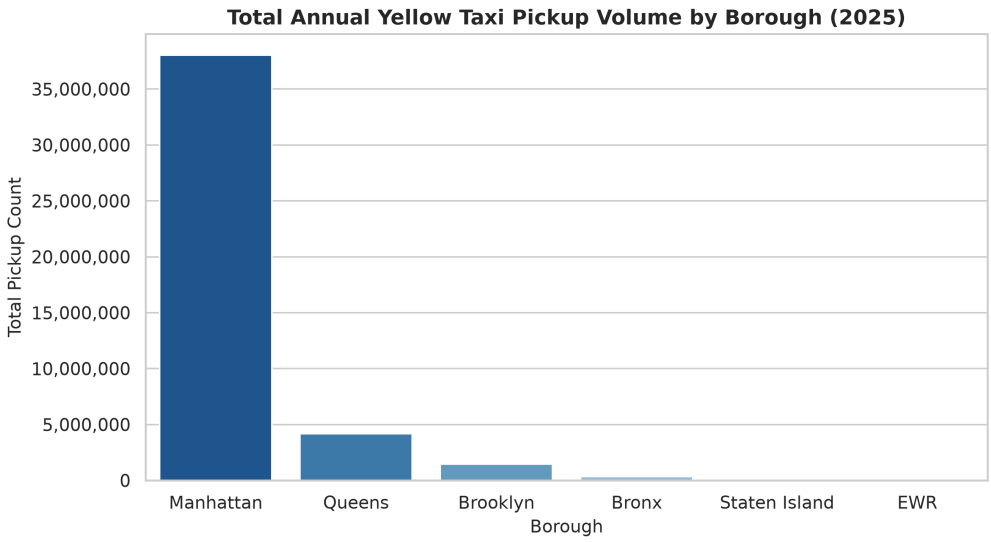
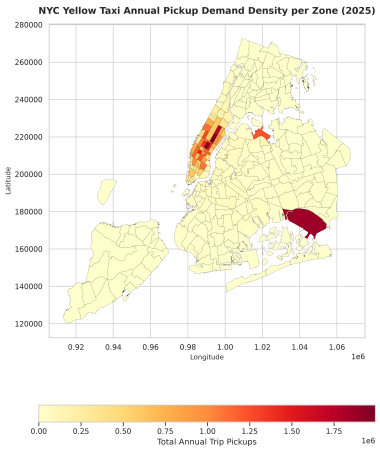


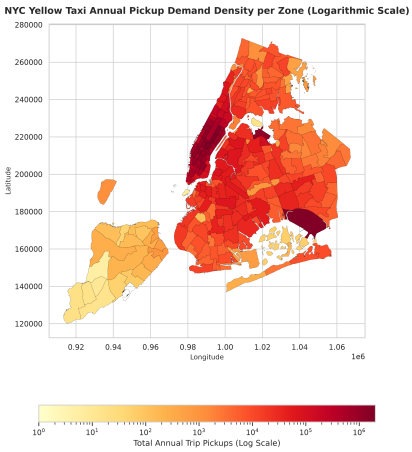
Figure 12: Comparison of Pickup and Drop-off Spatial Distributions across NYC Boroughs (2025).

5.3 Spatial Density Concentration: Linear vs. Logarithmic Analysis

Analyzing annual trip pick-up counts at the individual zone level further confirms the extreme spatial concentration observed in the borough totals.



(a) Linear Scale (Highlighting Extreme Core Concentration)



(b) Logarithmic Scale (Illuminating Outer-Borough Variance)

Figure 13: Annual Pickup Demand Density per Zone (2025).

As shown in Figure 13:

- **Linear Scale Map (Figure 13a):** Effectively captures extreme core hotspots exceeding 1.5×10^6 annual pickups in Midtown Manhattan and JFK Airport, but compresses outer-borough zones into a uniform low-density tier.
- **Logarithmic Scale Map (Figure 13b):** Successfully expands the lower-to-middle color spectrum (10^1 to 10^5 trips), revealing subtle variations, active commercial corridors in Brooklyn and Queens, and suburban transit patterns that would otherwise remain hidden behind extreme outliers.

5.4 Top Busiest Taxi Zones and Micro-Spatial Insights

To evaluate local demand hotspots, Table 20 lists the top 10 busiest pickup zones across New York City in 2025 alongside their corresponding average trip distances and fare amounts.

Table 20: Top 10 Busiest Taxi Pickup Zones in NYC (2025)

Zone Name	Borough	Pickup Count	Avg Distance (mi)	Avg Fare ($)
Upper East Side South	Manhattan	1,994,156	1.80	13.38
Midtown Center	Manhattan	1,942,432	2.34	17.04
JFK Airport	Queens	1,876,676	15.47	63.54
Upper East Side North	Manhattan	1,742,460	2.00	13.89
Penn Station/Madison Sq West	Manhattan	1,441,146	2.33	17.15
Midtown East	Manhattan	1,409,260	2.30	16.14
Times Sq/Theatre District	Manhattan	1,390,476	3.03	19.42
Lincoln Square East	Manhattan	1,283,771	2.21	14.72
LaGuardia Airport	Queens	1,223,332	9.61	45.54
Union Sq	Manhattan	1,185,896	2.12	15.20

Key analytical takeaways from the top 10 pickup zones include:

- **Residential & Commercial Core Dominance:** Upper East Side South and Midtown Center top the list with nearly 2 million pickups each. These Manhattan zones exhibit short average trip distances (~ 1.80–2.34 miles) and lower average fares ($\sim \$13.38$–$\$17.04$), reflecting dense intracity commuting.
- **Airport Hub Premiums:** JFK Airport and LaGuardia Airport rank among the busiest zones nationwide, generating over 3.1 million combined pickups. Unlike Manhattan zones, airport trips exhibit significantly longer trip distances (15.47 miles for JFK and 9.61 miles for LGA) and higher average fare amounts ($63.54 and $45.54 respectively), serving as high-yield revenue corridors.
- **Transit Hub Intersections:** Rail terminals and tourist hubs (Penn Station and Times Square) display elevated trip volumes with slightly higher average distances (up to 3.03 miles), acting as primary distribution points across Midtown Manhattan.

5.5 Borough-Level Descriptive and Distribution Analysis

To analyze traffic dynamics across different regional jurisdictions, trip metrics including trip distance, fare amount, trip duration, and calculated operational speed were examined on a borough-by-borough basis.

5.5.1 Manhattan

Manhattan represents the primary epicenter of yellow taxi activity, accounting for over 38 million pickup and drop-off trips. Table 21 summarizes the key parametric and non-parametric statistical metrics for Manhattan trips.

As visualized in Figure 14, the distributions across all four variables exhibit noticeable right-skewness:

- **Trip Distance & Fare Amount:** Intracity trips are predominantly short, with a median distance of 1.67 miles and a median fare of $\$13.28$. Standard deviations and upper quantiles reflect longer cross-borough trips extending into the tail.
- **Trip Duration & Speed:** Heavy traffic congestion across Manhattan's street grid restricts average vehicle movement, yielding a median trip duration of 12.34 minutes and a low median operational speed of 8.91 mph.

Table 21: Descriptive Statistics for Manhattan Trip Attributes (2025)

Spatial Type	Total Trips	Metric	Distance (mi)	Fare ($)	Duration (min)	Speed (mph)
Pickup	38,034,750	Mean	2.50	16.38	14.73	9.82
		Std	2.76	58.22	10.43	4.92
		Q1	1.02	9.30	7.77	6.70
		Median	1.67	13.28	12.34	8.91
		Q3	2.84	19.72	18.87	11.73
Dropoff	38,352,036	Mean	2.74	17.17	15.36	10.02
		Std	3.35	58.01	11.76	5.21
		Q1	1.02	9.27	7.81	6.73
		Median	1.68	13.43	12.45	8.95
		Q3	2.90	20.01	19.20	11.90

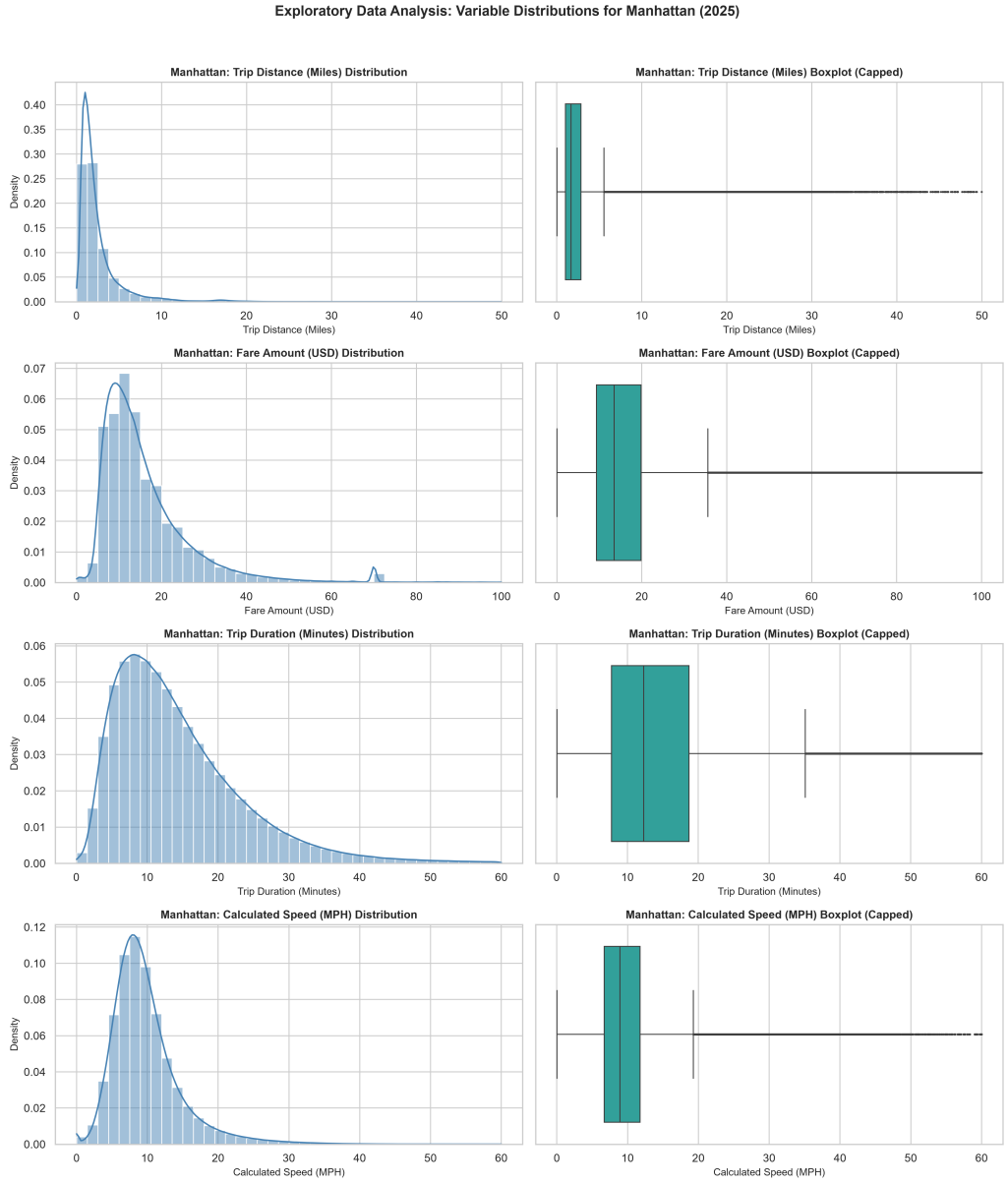


Figure 14: Distribution plots (Histograms with KDE) and boxplots for key trip variables in Manhattan (2025).

5.5.2 Queens

Queens exhibits a distinctly different operational profile compared to Manhattan, heavily influenced by expansive residential layouts and the presence of major aviation hubs (JFK and LaGuardia Airports). Across the dataset, Queens accounts for $4,165,160$ pickup trips and $2,434,066$ drop-off trips. Table 22 details the corresponding descriptive metrics.

Table 22: Descriptive Statistics for Queens Trip Attributes (2025)

Spatial Type	Total Trips	Metric	Distance (mi)	Fare ($)	Duration (min)	Speed (mph)
Pickup	4,165,160	Mean	11.51	50.33	35.26	20.85
		Std	6.54	452.60	20.36	8.49
		Q1	7.36	33.16	20.11	14.63
		Median	10.30	46.48	31.97	19.76
		Q3	16.98	70.00	47.15	26.04
Dropoff	2,434,066	Mean	8.52	39.08	28.18	19.32
		Std	5.42	591.77	18.99	8.61
		Q1	4.14	21.96	15.31	12.95
		Median	8.11	36.12	23.93	18.00
		Q3	11.56	51.28	36.39	24.44

As illustrated in Figure 15, trip lengths and fares in Queens skew significantly higher due to long-distance airport transits:

- **Trip Distance & Fare Amount:** Pickup trips originating in Queens show a high median distance of 10.30 miles and a median fare of $\$46.48$, driven by long trips to or from airport terminals and neighboring boroughs. Drop-off trips are somewhat shorter, with a median distance of 8.11 miles and a median fare of $\$36.12$.
- **Trip Duration & Speed:** Due to longer stretches on expressways and highways (such as the Van Wyck and Grand Central Parkway), vehicle speeds are considerably higher than in Manhattan, recording a median pickup speed of 19.76 mph and a median duration of 31.97 minutes.

5.5.3 Brooklyn

Brooklyn exhibits a distinct suburban-urban mix characterized by strong inter-borough connectivity and extensive residential trip generation, accounting for $1,471,290$ pickup trips and $2,455,952$ drop-off trips. Table 23 summarizes the key parametric and non-parametric statistical metrics for Brooklyn trips.

Table 23: Descriptive Statistics for Brooklyn Trip Attributes (2025 Sampled Data)

Spatial Type	Total Trips	Metric	Distance (mi)	Fare ($)	Duration (min)	Speed (mph)
Pickup	1,471,290	Mean	5.57	26.85	27.36	12.53
		Std	4.42	15.50	19.07	6.43
		Q1	2.53	16.87	15.15	8.48
		Median	4.29	24.39	22.80	11.11
		Q3	7.33	34.91	33.55	15.02
Dropoff	2,455,952	Mean	7.27	34.07	29.89	14.70
		Std	5.27	19.43	17.73	7.50
		Q1	3.37	20.43	17.83	9.71
		Median	5.91	30.24	26.26	12.89
		Q3	10.05	43.96	37.64	17.95

As visualized in Figure 16, the distributions across all four variables exhibit noticeable right-skewness:

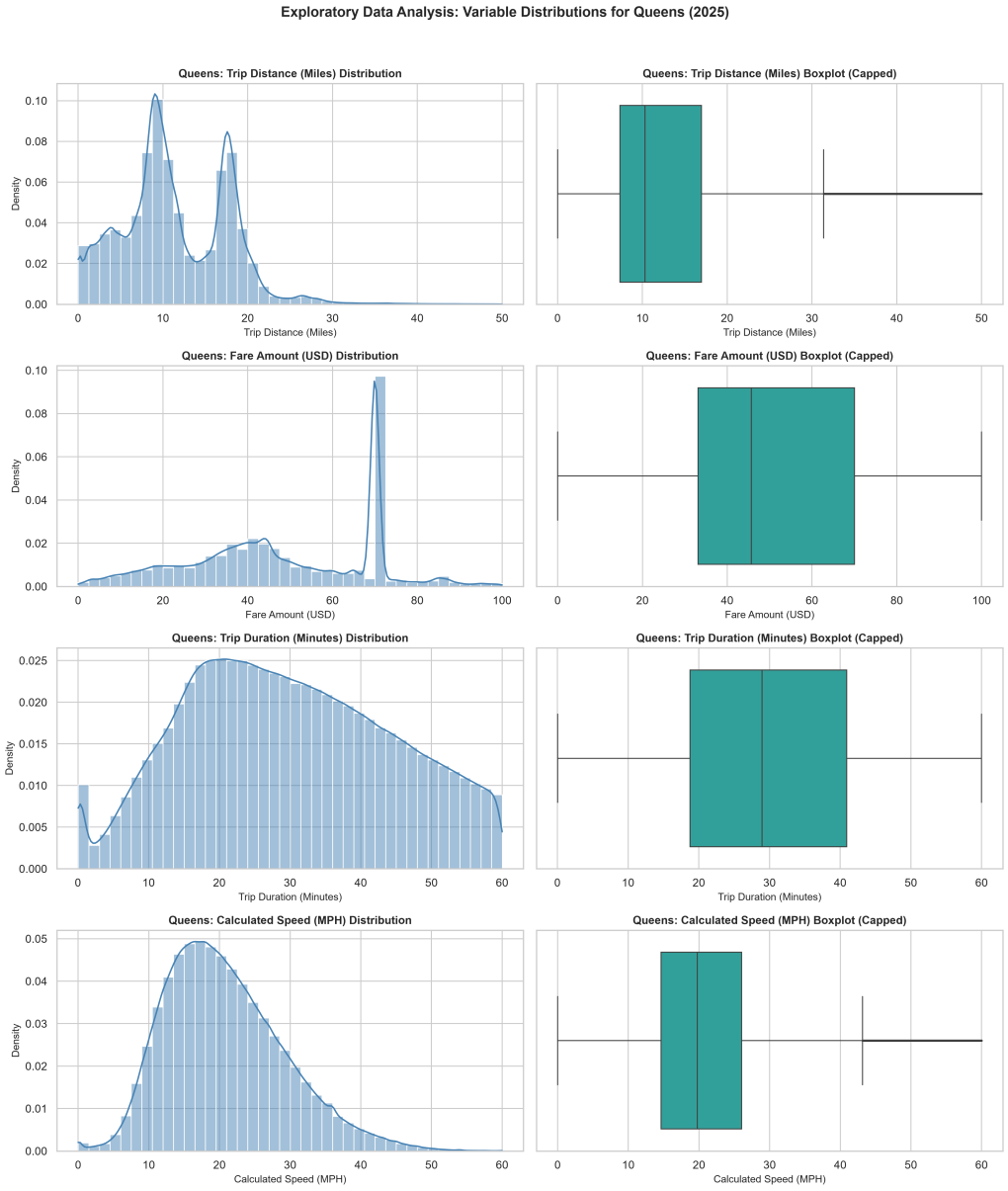


Figure 15: Distribution plots (Histograms with KDE) and boxplots for key trip variables in Queens (2025).

- **Trip Distance & Fare Amount:** Trips originating in or arriving at Brooklyn are longer on average than Manhattan trips, with median pickup distances of 4.29 miles and median fares of $24.39. Drop-offs are even more extended (median distance of 5.91 miles and median fare of $30.24), reflecting cross-river travel connecting Manhattan with Brooklyn commercial and residential hubs.
- **Trip Duration & Speed:** Operational speeds are moderately higher than in Manhattan due to broader thoroughfares and less dense gridlock, yielding a median pickup speed of 11.11 mph and a median duration of 22.80 minutes.

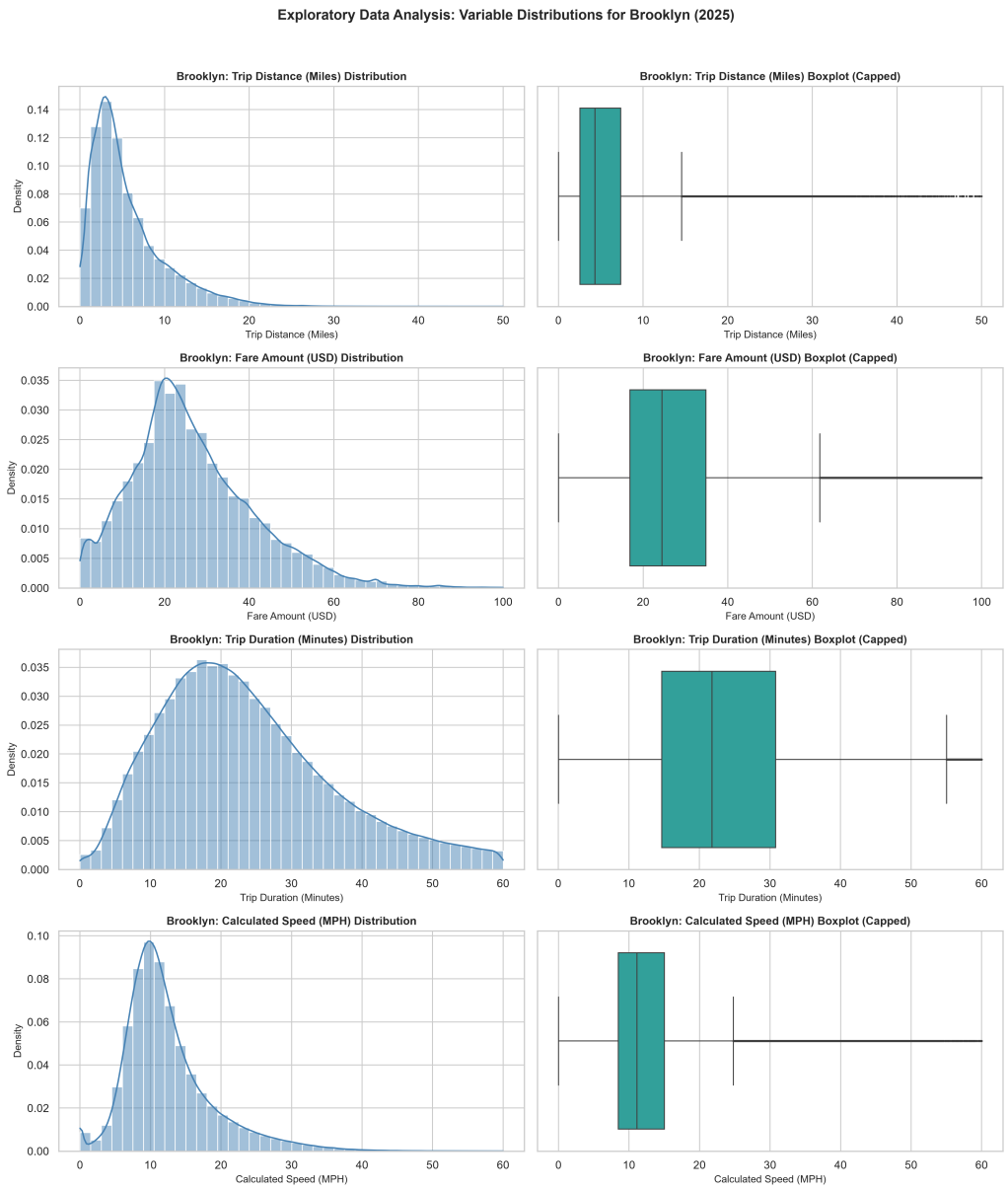


Figure 16: Distribution plots (Histograms with KDE) and boxplots for key trip variables in Brooklyn (2025).

5.5.4 Bronx

The Bronx represents a distinct outer-borough operational environment characterized by moderate local trip generation and longer cross-border commuting patterns, accounting for 339, 103 pickup trips and

493, 819 drop-off trips. Table 24 summarizes the key parametric and non-parametric statistical metrics for Bronx trips.

Table 24: Descriptive Statistics for Bronx Trip Attributes (2025 Sampled Data)

Spatial Type	Total Trips	Metric	Distance (mi)	Fare ($)	Duration (min)	Speed (mph)
Pickup	339,103	Mean	7.13	28.64	31.30	14.76
		Std	5.29	15.96	24.07	7.55
		Q1	2.96	17.48	14.73	9.34
		Median	5.85	26.54	23.34	13.04
		Q3	10.22	38.48	41.00	18.94
Dropoff	493,819	Mean	9.02	35.79	31.45	18.29
		Std	5.91	20.47	21.37	8.98
		Q1	4.25	21.20	17.37	11.05
		Median	8.30	33.19	25.93	17.21
		Q3	12.47	46.40	38.98	24.46

As visualized in Figure 17, the distributions across all four variables display pronounced right-skewness:

- **Trip Distance & Fare Amount:** Pickup trips originating in the Bronx show a median distance of 5.85 miles and a median fare of $26.54, while drop-offs extend further with a median distance of 8.30 miles and a median fare of $33.19, reflecting frequent inter-borough commuting to and from Manhattan.
- **Trip Duration & Speed:** Due to highway access and less restricted arterial networks relative to core commercial zones, operational speeds are higher, featuring a median pickup speed of 13.04 mph and a median drop-off speed of 17.21 mph.

5.5.5 Staten Island

Staten Island represents the most peripheral and lowest-volume operational sector for yellow taxi activity, accounting for only 3, 660 pickup trips and 11, 518 drop-off trips. Table 25 summarizes the key parametric and non-parametric statistical metrics for Staten Island trips.

Table 25: Descriptive Statistics for Staten Island Trip Attributes (2025)

Spatial Type	Total Trips	Metric	Distance (mi)	Fare ($)	Duration (min)	Speed (mph)
Pickup	3,660	Mean	9.47	32.76	26.57	22.76
		Std	6.98	29.37	19.29	9.73
		Q1	4.45	10.97	13.34	15.50
		Median	8.00	27.34	22.42	21.53
		Q3	13.25	49.07	35.93	28.54
Dropoff	11,518	Mean	19.52	74.87	44.38	27.76
		Std	8.38	36.56	21.93	8.97
		Q1	14.74	54.37	32.28	21.75
		Median	20.58	80.88	42.72	27.36
		Q3	25.56	98.39	55.15	33.67

As visualized in Figure 18, the distributions across all four variables exhibit unique patterns driven by long cross-borough commutes:

- **Trip Distance & Fare Amount:** While local pickups show moderate length (median distance of 8.00 miles and median fare of $27.34), drop-off trips terminating in Staten Island are exceptionally

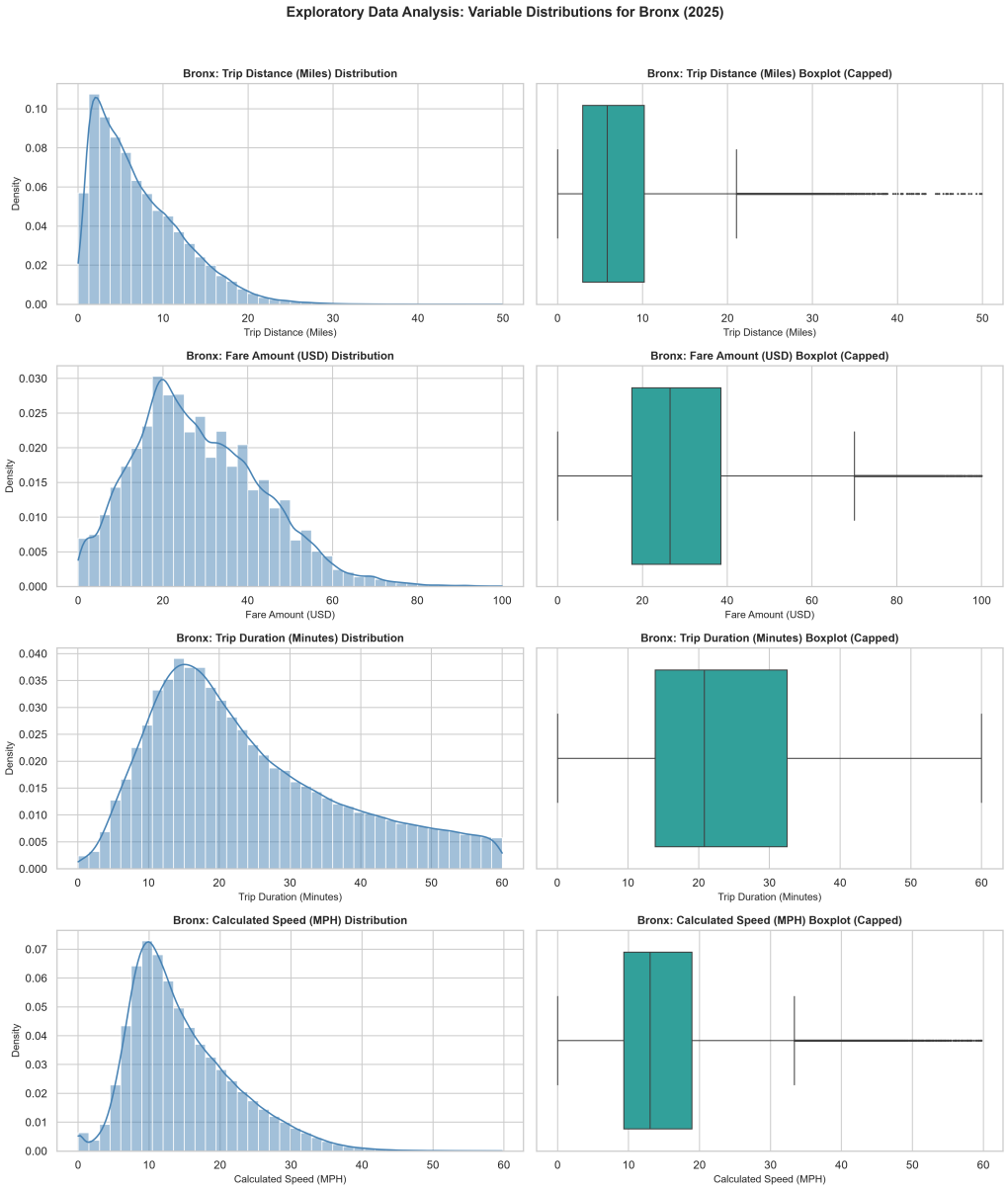


Figure 17: Distribution plots (Histograms with KDE) and boxplots for key trip variables in the Bronx (2025).

long and costly, showing a median distance of 20.58 miles and a median fare of $80.88. This reflects transits originating primarily from Manhattan or Brooklyn across bridges and expressways.

- **Trip Duration & Speed:** Due to heavy reliance on major highways and bridge crossings with limited local street congestion, operational speeds are significantly higher than inner-city boroughs, recording a median pickup speed of 21.53 mph and a median drop-off speed of 27.36 mph.

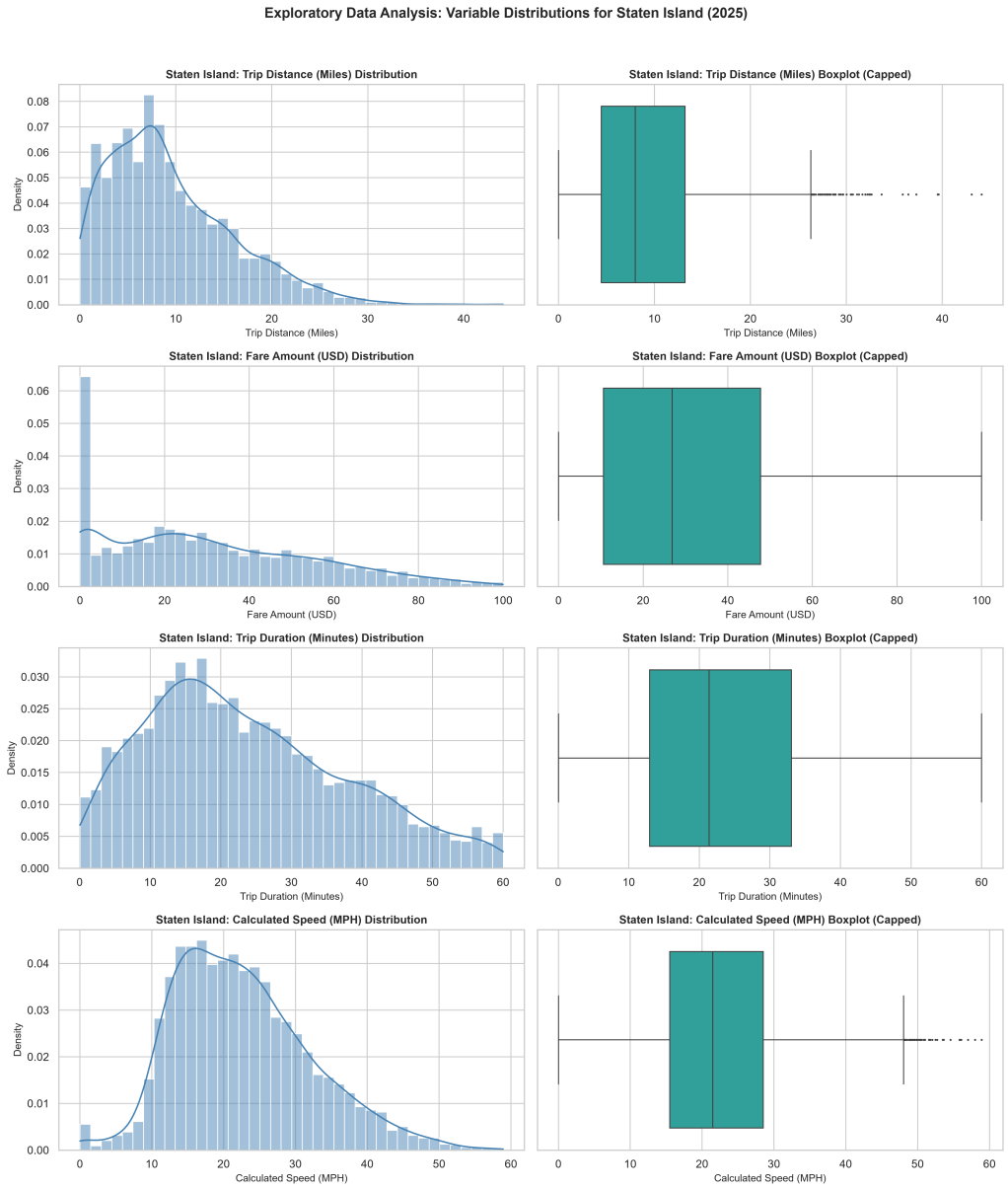


Figure 18: Distribution plots (Histograms with KDE) and boxplots for key trip variables in Staten Island (2025).

5.5.6 EWR

EWR (Newark Liberty International Airport) represents a specialized, highly asymmetric transportation terminal hub located outside New York City proper, accounting for 1,245 pickup trips and 84,257 drop-off trips. Table 26 summarizes the key parametric and non-parametric statistical metrics for EWR trips.

As visualized in Figure 19, trip attributes display distinct characteristics reflecting airport drop-off and minor local pick-up anomalies:

Table 26: Descriptive Statistics for EWR Trip Attributes (2025 Sampled Data)

Spatial Type	Total Trips	Metric	Distance (mi)	Fare ($)	Duration (min)	Speed (mph)
Pickup	1,245	Mean	3.86	84.36	9.27	21.97
		Std	8.17	44.55	19.61	12.27
		Q1	0.03	47.06	0.10	11.50
		Median	0.06	95.32	0.22	22.94
		Q3	1.15	115.13	3.59	30.07
Dropoff	84,257	Mean	17.85	97.56	41.52	28.04
		Std	4.90	20.95	16.63	8.63
		Q1	16.00	85.82	30.95	22.33
		Median	17.59	93.34	37.64	28.54
		Q3	19.45	103.51	48.23	34.10

- **Trip Distance & Fare Amount:** Drop-off trips arriving at EWR involve substantial inter-state or cross-borough travel, showing a high median distance of 17.59 miles and an elevated median fare of $\$93.34$. In contrast, pickups at EWR show near-zero median distances and durations due to localized terminal reassignment artifacts, though their flat or minimum-charge airport toll structures drive high median fares ($\$95.32$).
- **Trip Duration & Speed:** Highway transit via the New Jersey Turnpike network yields high operational speeds, with a median drop-off speed of 28.54 mph and a median pickup speed of 22.94 mph.

5.6 Summary Implications for Machine Learning Modeling

The structural disparity and extreme spatial concentration underscore the necessity of incorporating zone-level spatial embeddings, borough cross-features, and log-transformed demand metrics when developing predictive hourly demand models in subsequent project phases.

5.7 Multivariate Correlation and Pairwise Scatter Plot Analysis

To examine the linear dependencies and joint distributions among continuous trip attributes—namely **trip distance**, **fare amount**, **trip duration**, and **calculated speed**—multivariate correlation matrices and pairwise scatter plot matrices were generated for both the raw and cleaned datasets. Comparing these two states illustrates the profound distorting impact of uncleaned anomalies and data entry errors on statistical measures.

5.7.1 Analysis of Cleaned Data

In the cleaned dataset, where erroneous records, negative values, and physical impossibilities are filtered out, the correlation matrix and pairwise scatter plots reveal robust and physically coherent relationships:

- **Trip Distance and Duration ($r = 0.782$):** Exhibiting a strong positive correlation, longer trips naturally require more travel time. The pairwise scatter plot demonstrates a dense cluster around urban distances (< 10 miles) with a well-defined upward trend.
- **Trip Distance and Speed ($r = 0.688$):** A strong positive association emerges because longer trips frequently leverage expressways and highways (such as airport runs), allowing for higher average operational speeds.
- **Trip Duration and Speed ($r = 0.226$):** A moderate positive correlation indicates that longer-duration trips are often associated with high-speed highway or inter-borough transit rather than purely localized gridlock.

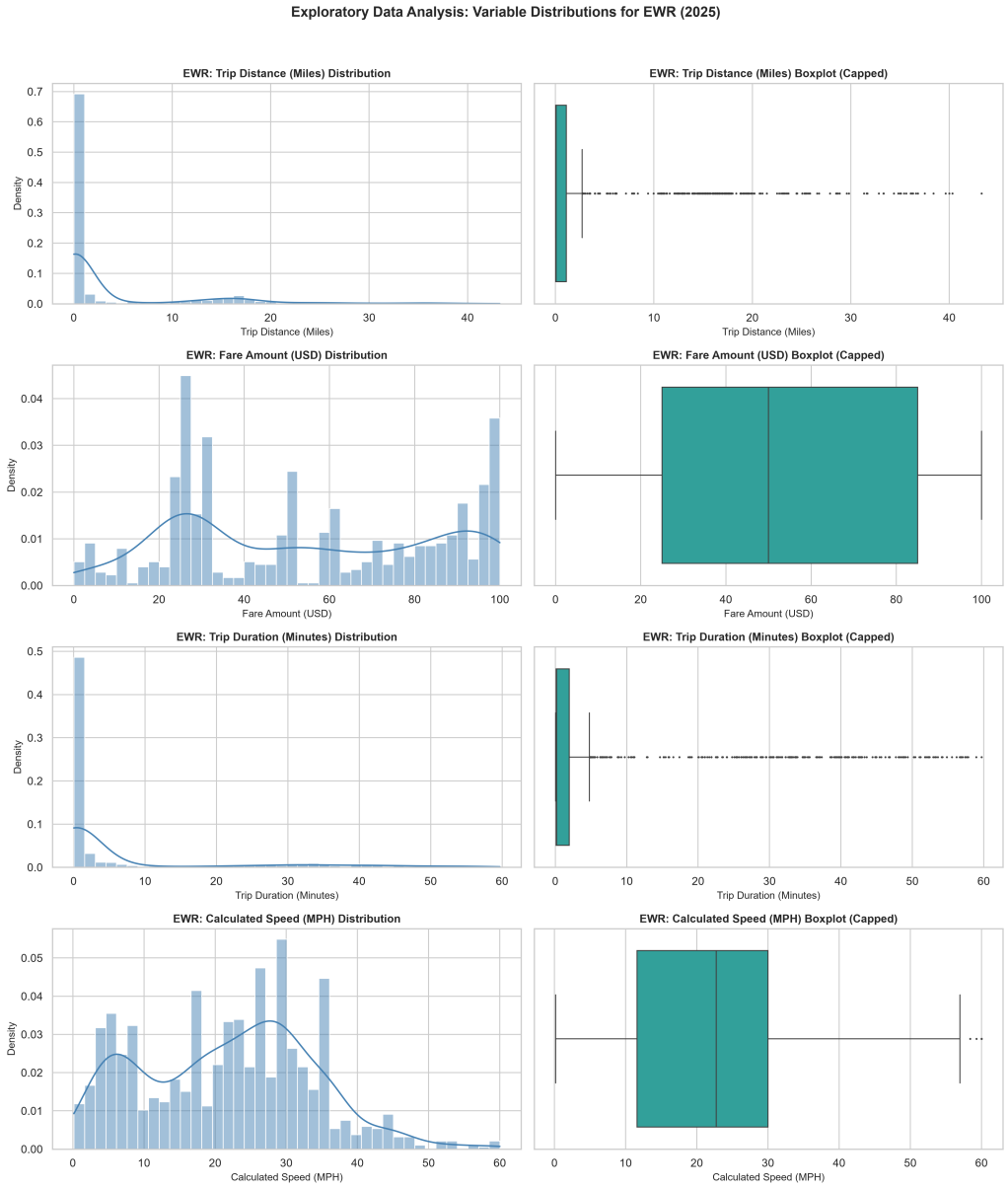


Figure 19: Distribution plots (Histograms with KDE) and boxplots for key trip variables in EWR (2025 Sampled Data).

- **Fare Amount Relationships:** While fare correlates intuitively with distance and duration, complex tariff structures (including base fares, tolls, surcharges, and tips) introduce dispersion, reflected in more moderate linear correlation coefficients ($r = 0.105$ with distance and $r = 0.092$ with duration).

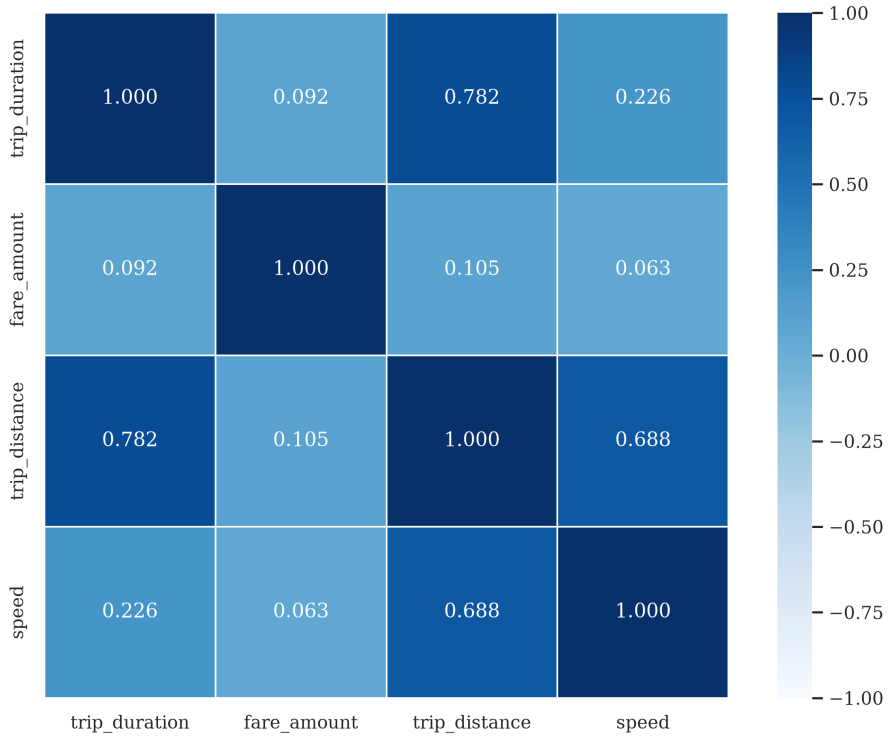


Figure 20: Pearson correlation matrix heatmap for trip attributes on the cleaned dataset.

5.7.2 Analysis of Raw Data and the Impact of Outliers

Conversely, the analysis of the raw, unfiltered dataset highlights the severe vulnerability of uncleaned administrative data to extreme outliers and data entry errors:

- **Distortion of Correlation Coefficients:** In the raw correlation matrix, relationships between variables collapse or approach zero (e.g., distance vs. duration drops to $r = 0.003$, and fare vs. distance drops to $r = 0.001$). This occurs because extreme magnitude outliers—such as recorded distances exceeding $19,000$ miles or speeds reaching $60,000$ mph—overwhelm the covariance structure.
- **Pairwise Compression:** As visualized in the raw pairwise scatter matrix (Figure 23), extreme outliers stretch the plot axes to absurd scales, compressing the entire legitimate population of millions of NYC yellow cab trips into an unreadable, microscopic point near the graph origin.

This comparison empirically underscores the absolute necessity of rigorous Phase 1 data cleaning protocols prior to conducting multivariate statistical modeling or econometric analysis.

6 Research Questions and Track Selection

To structure our investigation under **Track A: NYC TLC Structured Time-Series Analysis**, we formulate three primary analytical questions based on our exploratory data audit:

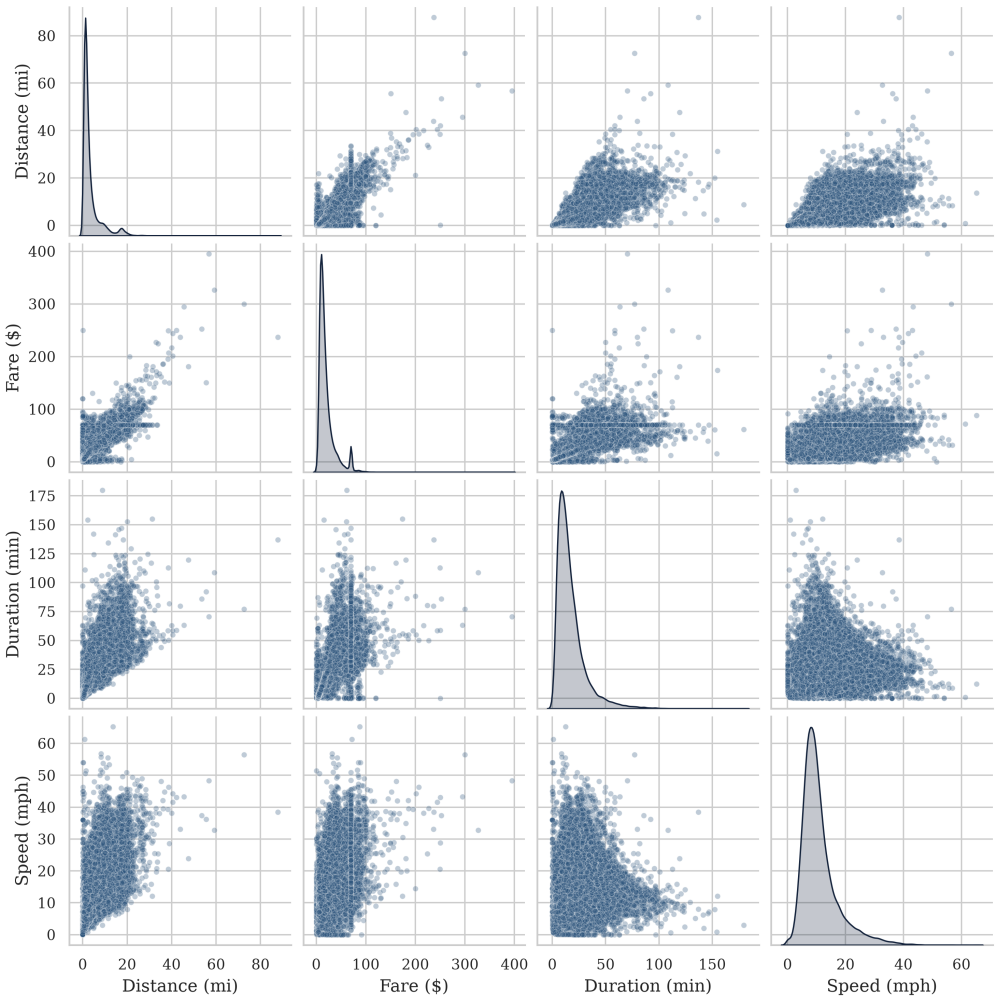


Figure 21: Pairwise scatter plot matrix and kernel density estimations for trip attributes on the cleaned dataset.

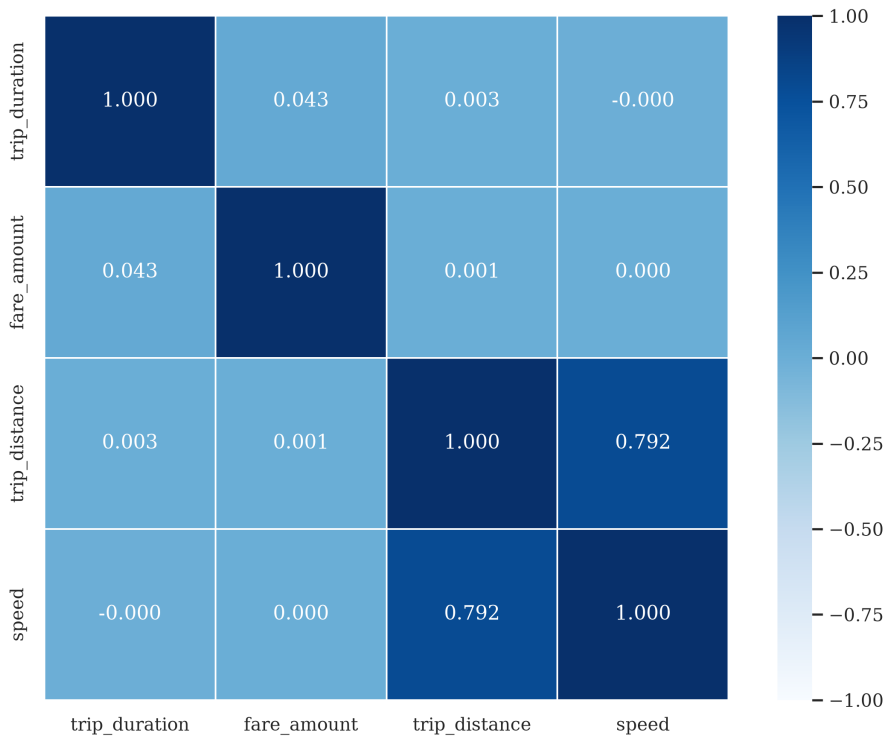


Figure 22: Pearson correlation matrix heatmap for trip attributes on the raw (unfiltered) dataset, showing severe attenuation due to outliers.

1. **Spatial Distribution Question:** How does trip density vary across official TLC zones, and what differentiates core commercial activity from outer-borough and airport travel?
2. **Financial & Policy Impact Question:** How do regulatory tolls (such as the 2025 CBD congestion fee) and airport flat rates influence fleet revenue and financial distributions?
3. **Time-Series Demand Forecasting Question:** *How accurately can we forecast hourly passenger trip originations per taxi zone $(z, t) \in$ Zone $\times$ Hour using cyclical temporal encodings, spatial features, and gradient-boosted decision trees under a strict temporal train/test split, and how do non-linear models compare against historical rolling-mean baselines?*

Phase 1 data auditing and spatial exploration confirm that the third question serves as our primary predictive objective, leveraging the spatio-temporal aggregation unit (z, t) established across the full calendar year 2025.

7 Methods

7.1 Aggregation and Problem Formulation

Let $z \in \{1, 2, \ldots, 263\}$ denote a TLC taxi zone, and t denote a discrete 1-hour time index. Raw trip records are aggregated into a spatio-temporal matrix:

$$y_{z,t} = \sum_{i=1}^{N} \mathbb{I}\left(\text{PULocationID}_i = z \ \wedge \ \lfloor \text{tpep_pickup_datetime}_i \rfloor = t\right) \tag{1}$$

where $y_{z,t} \in \mathbb{N}_0$ represents the target variable `trip_count`.

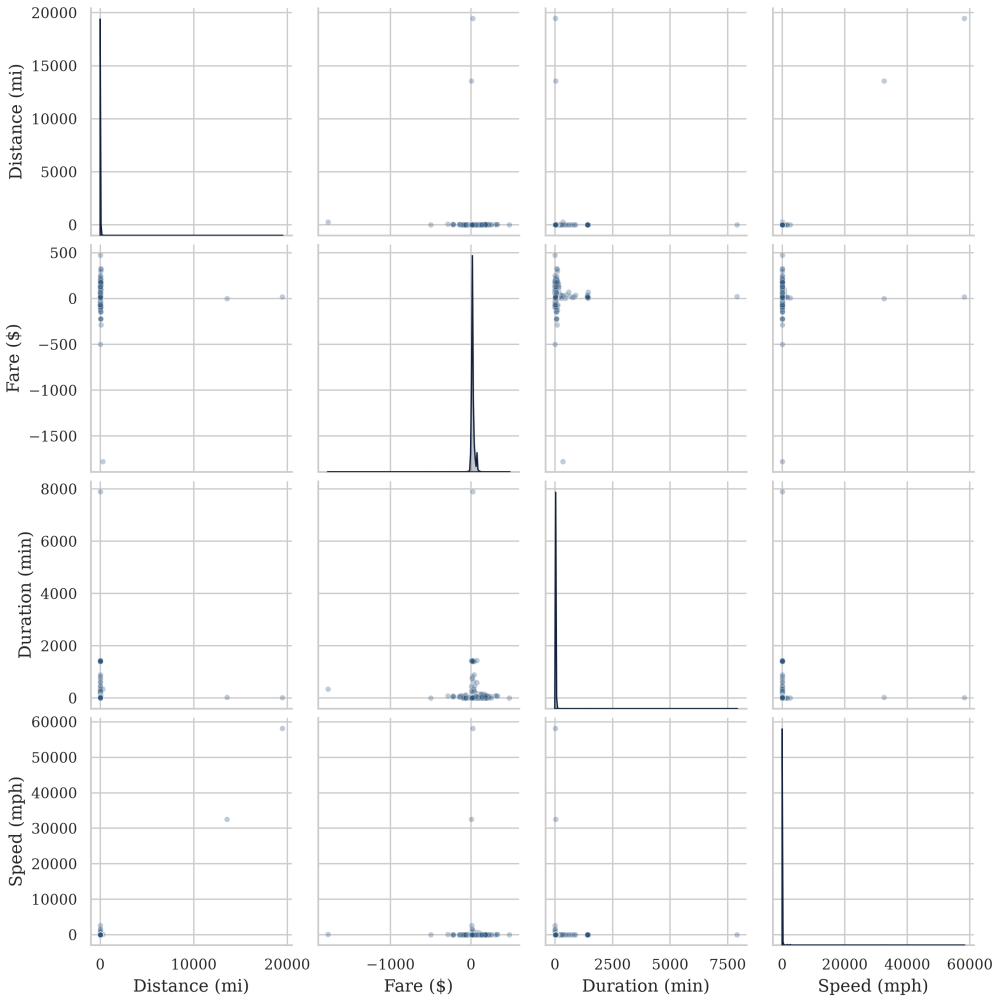


Figure 23: Pairwise scatter plot matrix for trip attributes on the raw dataset, illustrating axis distortion caused by extreme uncleaned anomalies.

8 Experimental Design

8.1 Strict Temporal Split

To prevent data leakage, evaluation is conducted strictly along the temporal axis:

- **Training Set:** January 1, 2025 – September 30, 2025 (Months 1–9, $\approx 75\%$ of temporal duration).
- **Validation / Test Set:** October 1, 2025 – December 31, 2025 (Months 10–12, $\approx 25\%$ of temporal duration).

8.2 Evaluation Metrics

Models will be evaluated across all spatial zones using Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE):

$$\text{RMSE} = \sqrt{\frac{1}{Z \times T} \sum_{z=1}^{Z} \sum_{t=1}^{T} \left(y_{z,t} - \hat{y}_{z,t}\right)^2} \tag{2}$$

$$\text{MAE} = \frac{1}{Z \times T} \sum_{z=1}^{Z} \sum_{t=1}^{T} \left|y_{z,t} - \hat{y}_{z,t}\right| \tag{3}$$

9 Results

Execution of the DuckDB analytical engine across all 12 monthly Parquet files successfully audited $48,722,602$ raw records. Applying our systematic cleansing protocols successfully removed $4,707,394$ anomalous records ($\approx 9.66\%$ noise rate), yielding a pristine, high-fidelity baseline population of $44,015,208$ **valid trips** dedicated for downstream spatio-temporal aggregation.

Key quantitative findings from our Phase 1 investigation include:

- **Spatial Concentration:** Trip originations exhibit extreme spatial concentration. Manhattan dominates fleet activity, led by Upper East Side South ($\approx$ 1.99M pickups) and Midtown Center (≈ 1.94M pickups). Major aviation hubs (JFK and LaGuardia Airports) generate over 3.1M combined pickups, serving as high-yield regional corridors characterized by longer distances (~ 15.47 miles for JFK) and higher average fares ($\sim \$63.54$).
- **Temporal Rhythms:** Hourly demand follows a robust bimodal commuter curve peaking at 18:00 ($\approx$ 3.5M annual originations), with weekly volumes peaking on Saturdays ($\approx$ 7.5M trips) and monthly volume peaking in May (≈ 4.6M trips).
- **Financial and Operational Metrics:** Cleansing negative fares ($-\$3$ to $-\$24$) and extreme GPS telemetry errors stabilized core operational metrics, establishing a clean median trip distance of 1.89 miles, a median fare of $\$14.20$, and a median urban speed of 9.43 mph.

These empirical insights directly validate our spatio-temporal aggregation unit (z, t) and confirm that the selected demand forecasting question is fully supported by the cleaned dataset.

10 Data Leakage and Prevention Strategy

In spatio-temporal forecasting and time-series modeling, data leakage represents a critical risk where information from outside the training period or future target observations inadvertently contaminates the training set, leading to overly optimistic and ungeneralizable performance metrics. To ensure that our evaluation design realistically simulates production forecasting, we explicitly identify and mitigate three primary vectors of potential data leakage:

- **Temporal Look-Ahead Bias:** Random shuffling or standard k-fold cross-validation mixes past and future observations, allowing models to "see around corners" using future traffic states to predict past demand. *Mitigation:* We enforce a strict chronological temporal split (Training on January–September 2025; Validation/Testing on October–December 2025). The model never accesses information from periods subsequent to the training window.
- **Global Aggregation Leakage:** Computing global summary statistics (such as means, standard deviations, or min-max scaling bounds) across the entire 12-month annual dataset prior to partitioning introduces future distribution parameters into the training phase. *Mitigation:* All feature engineering transformations, rolling averages, and lag calculations are computed exclusively using rolling windows or historical parameters derived strictly from the training partition ($t \leq$ September 30, 2025).
- **Spatial Target Leakage via Rolling Lags:** When constructing spatial lag features or historical moving averages across adjacent taxi zones z, improper window definitions can inadvertently incorporate contemporaneous hour targets (t) rather than lagged history ($t-1, t-2, \dots$). *Mitigation:* All autoregressive lag features ($y_{z,t-k}$) are strictly lagged by at least one full hour ($k \geq 1$) to ensure that predictions for time index t rely solely on historically realized information available at the time of inference.

By adhering to these rigorous partitioning and feature-engineering protocols, we eliminate structural leakage risks and guarantee that our validation scores reflect true out-of-sample generalization.

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11 Robustness / Sensitivity / Shift Analysis

The key temporal shift in 2025 stems from evolving urban mobility patterns and seasonal commuter variations. Future phases will evaluate whether model residuals increase in high-density commercial core clusters during peak transitional months compared to outer borough controls.

12 Ethics, Privacy and Governance

The dataset contains no Personally Identifiable Information (PII). Vendor IDs are coarse integer codes, and spatial locations are masked at the Taxi Zone level (265 spatial polygons), preventing individual driver or passenger tracking.

13 Limitations

External environmental factors (e.g., hourly rainfall, snowfall, subway service disruptions) are not present in the native TLC dataset and will require external data ingestion in Phase 2.

14 Conclusion

Phase 1 successfully establishes a rigorous analytical pipeline, quantifies data quality boundaries, and formalizes the prediction task under **Track A: NYC TLC Structured Time-Series Analysis**. By integrating high-performance DuckDB data auditing, spatial GIS mapping across official NYC taxi zones, and comprehensive exploratory data analysis, we have resolved critical telemetry anomalies while preserving authentic economic and transport activity.

With the research questions defined and the clean baseline population of 44.0M trips established, Phase 2 will proceed to spatio-temporal feature engineering—incorporating cyclical temporal encodings,

spatial lag features, and rolling-mean baselines—to train and evaluate gradient-boosted decision trees under our strict temporal train/test split.

15 References

1. NYC Taxi and Limousine Commission, *TLC Trip Record Data (2025)*, official NYC Open Data portal.
2. DuckDB Development Team, *DuckDB: An In-Process Analytical Database Management System*, 2026.

16 Supplementary Material

Data audit and exploratory analysis are executable via the project repository files:

- **Jupyter Notebooks (`notebooks/`):**
  - `notebooks/data_audit.ipynb`
  - `notebooks/data_cleaning_pipeline.ipynb`
  - `notebooks/eda_001.ipynb` through `notebooks/eda_004.ipynb`
- **Data Ingestion Scripts (`src/`):**
  - `src/download_data.py`
  - `src/download_spatial.py`

Note: For further execution instructions, please refer to the `README.md` file in the project's source code repository at https://github.com/CDsonji/NYC-TLC-Trip-Record-Data-Analysis-.git.